DUALITY IN DECONFOUNDING WITH INSTRUMENTS AND SURROGATE CONTROLS

Haoran Lei and Jin-Hong Du

Department of Statistics and Actuarial Science, The University of Hong Kong

Abstract:

Key words and phrases:

1. Introduction

Designed perturbations and auxiliary measurements provide opportunities to address unmeasured confounding. However, randomization of a perturbation does not ensure exogeneity of the resulting exposure. We study how instruments and surrogate controls can be used separately and jointly to estimate linear structural effects.

Single-cell Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) perturbation studies provide a motivating example. For a fixed target gene g, let z_i denote the assigned single-guide RNA (sgRNA) or guide-pair indicator, x_i the measured expression of the target gene, and

y_i the expression of a candidate downstream gene. Although z_i is experimentally assigned, regression of y_i on x_i may remain confounded by latent cellular and technical factors, so perturbation randomization does not imply exogeneity of x_i. Here, z_i provides instrument-like variation in x_i, while surrogate controls w_i, constructed for example from negative-control or housekeeping genes, may capture latent confounding. These sources are complementary: instruments may be low-dimensional or weak, whereas surrogate controls may provide richer confounding information.

We study a linear framework in which instrumental-variable (IV) and surrogate-control (SV) methods generate two projected moment restrictions for the same structural parameter. In the scalar-outcome model $y_i = x_i^\top \beta + \varsigma_i$, the IV channel uses $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$, implied by $\mathbb{E}(z_i\varsigma_i) = 0_q$, whereas the SV channel uses $\mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} = 0_p$. Geometrically, IV retains the component of x_i explained by z_i, whereas SV retains the component orthogonal to the span of w_i. Both yield p-dimensional projected moment blocks, providing a common framework for identification, first-order behavior, joint weighting, and sensitivity to misspecification. The construction extends columnwise to multivariate outcomes $Y = XB + \Xi$.

Our framework connects several strands of the literature. Classical instrumental-variable methods address endogeneity through relevance and

orthogonality restrictions, with 2SLS and GMM providing standard projection- and moment-based approaches (Anderson and Rubin, 1949; Theil, 1953; Sargan, 1958; Hansen, 1982). Our projected IV formulation uses the p-dimensional component of the classical q-dimensional IV moment relevant for projecting x_i. Related proxy-variable, negative-control, proximal, and latent-factor methods use auxiliary measurements to address latent confounding (Miao et al., 2018; Shi et al., 2020; Miao et al., 2024; Tchetgen Tchetgen et al., 2024; Wang and Blei, 2019; Adams and Hansen, 2025). Particularly close is Miao et al. (2023), which unifies IV-like and proxy-like information in a nonparametric identification framework; we instead focus on a linear structural coefficient and the projection, rank, first-order, and weighting structure of the resulting moments. Our framework is also related to recent work combining multiple deconfounding signals (Dukes et al., 2025; Imbens et al., 2025; Kallus and Mao, 2025). Rather than recovering the latent confounder, we treat z_i and w_i as observed ancillary signals and study their projected moments under exact and locally misspecified restrictions.

Our contributions are threefold. First, we place IV and SV adjustment in a common p-dimensional projected-moment representation based on $\Pi_z x_i$ and $\Pi_w^{\perp} x_i$, respectively, making explicit their shared structure and distinct

projection geometries. Second, for each single-source channel, we characterize the statistical target under projected-moment violations and establish identification, consistency, and asymptotic normality of the corresponding feasible estimator, while separating statistical identification from the additional assumptions required for causal interpretation (Theorems 1 and 2). Third, we combine the two projected blocks in a $2p$-dimensional GMM system whose stacked Jacobian can identify the coefficient even when an individual block is rank deficient (Theorem 4). We further characterize the information retained by projection, establish the efficiency properties of optimal combination under correct specification, and study the bias–variance tradeoff under $n^{-1/2}$-local misspecification through local asymptotic bias and worst-case asymptotic mean squared error (AMSE) (Propositions 7, 8, 11 and 12 and theorem 5).

2. Preliminaries

2.1 Ancillary variables for deconfounding

Consider the confounded linear regression model

$$y = x^\top \beta + \varsigma, \tag{2.1}$$

where $y \in \mathbb{R}$, $x \in \mathbb{R}^p$, $\beta \in \mathbb{R}^p$, and $\varsigma \in \mathbb{R}$. We assume $\mathbb{E}(\varsigma) = 0$, $\mathrm{Var}(\varsigma) = \sigma^2 < \infty$, and $\Sigma_x := \mathbb{E}(xx^\top) \in \mathbb{S}^p_{++}$. When $\mathbb{E}(x\varsigma) = 0$, ordinary least squares consistently estimates β. Our interest is instead in settings where $\mathbb{E}(x\varsigma) \neq 0$, for example because the disturbance contains a latent confounding component $u \in \mathbb{R}^r$ associated with both the regressors and the outcome; see Figure 1.

For n independent and identically distributed (i.i.d.) observations, write the sample model as

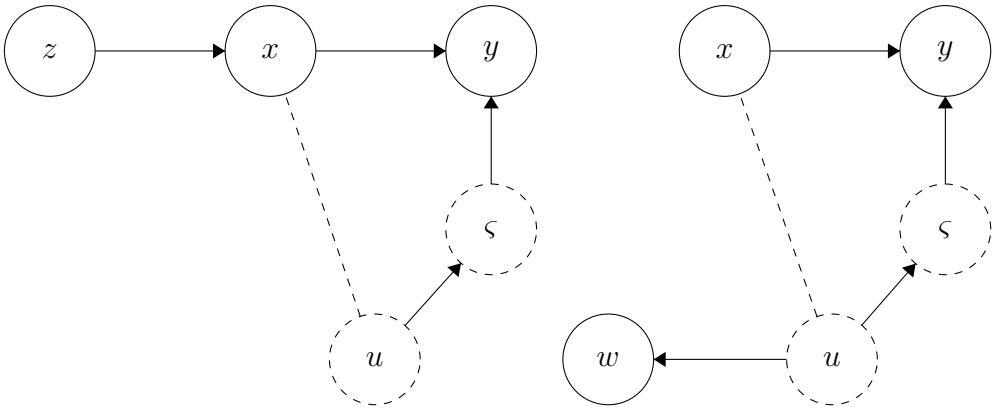
$$Y = X\beta + \varsigma, \tag{2.2}$$

where $Y = (y_1, \ldots, y_n)^\top \in \mathbb{R}^n$, $X = (x_1, \ldots, x_n)^\top \in \mathbb{R}^{n\times p}$, and $\varsigma = (\varsigma_1, \ldots, \varsigma_n)^\top \in \mathbb{R}^n$.

To address latent confounding, we use observed ancillary variables that provide information beyond x_i. The IV and SV channels exploit, respectively, instrument-induced and residualized variation in x_i. Their distinct projection geometries are illustrated in Figure 1 and developed below.

2.2 Instrumental variable models

Suppose, in addition, that an instrument $z_i \in \mathbb{R}^q$ is observed for each i. Under the centered formulation, let $\Sigma_z := \mathbb{E}(z_i z_i^\top)$, $\Sigma_{xz} := \mathbb{E}(x_i z_i^\top)$, and $\Sigma_{zx} := \Sigma_{xz}^\top$. When $\Sigma_z \in \mathbb{S}^q_{++}$, define the population projection $\Pi_z x_i :=$



(a) Instrumental variable **(b)** Surrogate control

Figure 1: Causal diagrams under (more restricted) conditional independence assumptions.

$\Sigma_{xz}\Sigma_z^{-1}z_i$.

Define the projected-moment bias $\delta_{\text{IV}} := \mathbb{E}\{(\Pi_z x_i)\varsigma_i\}$ and relevance matrix $Q_{\text{IV}} := \mathbb{E}\{(\Pi_z x_i)x_i^\top\} = \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx}$. Whenever $Q_{\text{IV}} \in \mathbb{S}_{++}^p$, the corresponding statistical target is $\beta_{\text{IV}} := \beta + Q_{\text{IV}}^{-1}\delta_{\text{IV}}$. Thus, valid IV, $\mathbb{E}(z_i\varsigma_i) = 0_q$, implies $\delta_{\text{IV}} = 0_p$ and $\beta_{\text{IV}} = \beta$, whereas $\delta_{\text{IV}} \neq 0_p$ represents projected IV misspecification.

At the sample level, let $Z = (z_1, \ldots, z_n)^\top \in \mathbb{R}^{n\times q}$ and $P_Z := Z(Z^\top Z)^{-1}Z^\top$. The first stage projects the regressors onto the instrument span, giving the fitted design $\widehat{X} := P_Z X$. Whenever $Z^\top Z$ and $X^\top P_Z X$ are nonsingular, the

second-stage estimator is

$$\widehat{\beta}_{\text{IV}} := \operatorname*{argmin}_{\theta \in \mathbb{R}^p} \|Y - \widehat{X}\theta\|_2^2 = (X^\top P_Z X)^{-1} X^\top P_Z Y. \tag{2.3}$$

Equivalently, the sample rank conditions are $\operatorname{rank}(Z) = q$ and $\operatorname{rank}(Z^\top X) = p$, which necessarily require $p \leq q \leq n$. Throughout the fixed-dimensional asymptotic analysis, p and q are treated as fixed as $n \to \infty$.

For the first-order distribution, define $e_{\text{IV},i} := y_i - x_i^\top \beta_{\text{IV}} - z_i^\top \Sigma_z^{-1} \mathbb{E}[z_i(y_i - x_i^\top \beta_{\text{IV}})]$, so that $\mathbb{E}(z_i e_{\text{IV},i}) = 0$. Further define $\psi_{\text{IV},i} := (\Pi_z x_i) e_{\text{IV},i} + x_i z_i^\top \Sigma_z^{-1} \mathbb{E}[z_i(y_i - x_i^\top \beta_{\text{IV}})]$ and $\Sigma_{\text{IV}} := \operatorname{Var}(\psi_{\text{IV},i})$.

Assumption 1 (Projected instrumental variables conditions)**.** Suppose the following conditions hold.

(i) *Projected relevance.* Suppose $\Sigma_z \in \mathbb{S}_{++}^q$ and $Q_{\text{IV}} \in \mathbb{S}_{++}^p$.

(ii) *Second moments and nondegeneracy.* Suppose $\mathbb{E}[\|x_i z_i^\top\|_{\text{op}}^2] < \infty$, $\mathbb{E}\|z_i e_{\text{IV},i}\|_2^2 < \infty$, and $\Sigma_{\text{IV}} \in \mathbb{S}_{++}^p$.

Theorem 1 (Identification and asymptotic properties of projected 2SLS)**.** Under Assumption 1(i), the following statements hold. (i) *Identification.* The projected population moment equation $\mathbb{E}[(\Pi_z x_i)(y_i - x_i^\top \theta)] = 0$ has the unique solution $\theta = \beta_{\text{IV}}$. (ii) *Consistency.* As $n \to \infty$, $\widehat{\beta}_{\text{IV}} \xrightarrow{\text{p}} \beta_{\text{IV}}$. (iii) *Asymptotic normality.* If, in addition, Assumption 1(ii) holds, then $\sqrt{n}(\widehat{\beta}_{\text{IV}} - \beta_{\text{IV}}) \xrightarrow{\text{d}} \mathcal{N}(0, V_{\text{IV}})$, where $V_{\text{IV}} := Q_{\text{IV}}^{-1} \Sigma_{\text{IV}} Q_{\text{IV}}^{-1}$.

Moreover, $\operatorname{rank}(Q_{\mathrm{IV}}) = \operatorname{rank}(\Sigma_{zx}) \leq \min(p, q)$, so IV-only identification requires $\operatorname{rank}(\Sigma_{zx}) = p$ and hence $q \geq p$, although the stacked IV–SV system may remain identifying when an individual block is rank deficient. When $q > p$, projection may discard overidentifying information; see Appendix A.1 for further discussion.

2.3 Surrogate controls and partialling out

Suppose, in addition, that surrogate controls $w_i \in \mathbb{R}^m$ are observed. Under the centered formulation, let $\Sigma_w := \mathbb{E}(w_i w_i^\top)$ and $\Sigma_{xw} := \mathbb{E}(x_i w_i^\top)$. When $\Sigma_w \in \mathbb{S}^m_{++}$, define $\Pi_w x_i := \Sigma_{xw}\Sigma_w^{-1} w_i$ and $\Pi_w^\perp x_i := x_i - \Pi_w x_i$. Unlike IV, SV retains the component of x_i orthogonal to the surrogate-control span. Exact validity requires $\mathbb{E}[(\Pi_w^\perp x_i)\varsigma_i] = 0_p$, without requiring $\mathbb{E}(x_i\varsigma_i) = 0_p$ or exogeneity of w_i.

Define the residualized-moment bias $\delta_{\mathrm{SV}} := \mathbb{E}[(\Pi_w^\perp x_i)\varsigma_i]$ and relevance matrix $Q_{\mathrm{SV}} := \mathbb{E}[(\Pi_w^\perp x_i)x_i^\top]$. Whenever $Q_{\mathrm{SV}} \in \mathbb{S}^p_{++}$, the corresponding statistical target is $\beta_{\mathrm{SV}} := \beta + Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}$. Hence, exact validity gives $\beta_{\mathrm{SV}} = \beta$, while $\delta_{\mathrm{SV}} \neq 0_p$ represents residual confounding after surrogate adjustment.

At the sample level, let $W = (w_1, \ldots, w_n)^\top \in \mathbb{R}^{n\times m}$, $P_W := W(W^\top W)^{-1}W^\top$, and $P_W^\perp := I_n - P_W$. The first step partials out the surrogate controls from both the regressors and the outcome, giving $\widetilde{X} := P_W^\perp X$ and $\widetilde{Y} := P_W^\perp Y$.

Whenever $W^\top W$ and $X^\top P_W^\perp X$ are nonsingular, the second-step estimator is

$$\widehat{\beta}_{\mathrm{SV}} := \operatorname*{argmin}_{\theta \in \mathbb{R}^p} \|\widetilde{Y} - \widetilde{X}\theta\|_2^2 = (X^\top P_W^\perp X)^{-1} X^\top P_W^\perp Y. \tag{2.4}$$

Equivalently, the sample rank conditions are $\operatorname{rank}(W) = m$ and $\operatorname{rank}(P_W^\perp X) = p$, requiring $m + p \leq n$; throughout, m and p are fixed as $n \to \infty$.

For the first-order distribution, define $e_{\mathrm{SV},i} := y_i - x_i^\top \beta_{\mathrm{SV}} - w_i^\top \Sigma_w^{-1} \mathbb{E}[w_i(y_i - x_i^\top \beta_{\mathrm{SV}})]$, so that $\mathbb{E}(w_i e_{\mathrm{SV},i}) = 0$. Further define $\psi_{\mathrm{SV},i} := (\Pi_w^\perp x_i) e_{\mathrm{SV},i}$ and $\Sigma_{\mathrm{SV}} := \operatorname{Var}(\psi_{\mathrm{SV},i})$.

Assumption 2 (Partialling-out conditions with observed controls)**.** Suppose the following conditions hold.

(i) *Residualized relevance.* Suppose $\Sigma_w \in \mathbb{S}_{++}^m$ and $Q_{\mathrm{SV}} \in \mathbb{S}_{++}^p$.

(ii) *Second moments and nondegeneracy.* Suppose $\mathbb{E}\|x_i \varsigma_i\|_2^2 < \infty$, $\mathbb{E}\|x_i x_i^\top\|_{\mathrm{op}}^2 < \infty$, $\mathbb{E}\|x_i w_i^\top\|_{\mathrm{op}}^2 < \infty$, $\mathbb{E}\|w_i e_{\mathrm{SV},i}\|_2^2 < \infty$, and $\Sigma_{\mathrm{SV}} \in \mathbb{S}_{++}^p$.

Theorem 2 (Identification and asymptotic properties of partialling out)**.** Under Assumption 2(i), the following statements hold. (i) *Identification.* The residualized population moment equation $\mathbb{E}[(\Pi_w^\perp x_i)(y_i - x_i^\top \theta)] = 0$ has the unique solution $\theta = \beta_{\mathrm{SV}}$. (ii) *Consistency.* As $n \to \infty$, $\widehat{\beta}_{\mathrm{SV}} \xrightarrow{\mathrm{p}} \beta_{\mathrm{SV}}$. (iii) *Asymptotic normality.* If, in addition, Assumption 2(ii) holds, then $\sqrt{n}(\widehat{\beta}_{\mathrm{SV}} - \beta_{\mathrm{SV}}) \xrightarrow{\mathrm{d}} \mathcal{N}(0, V_{\mathrm{SV}})$, where $V_{\mathrm{SV}} := Q_{\mathrm{SV}}^{-1} \Sigma_{\mathrm{SV}} Q_{\mathrm{SV}}^{-1}$.

Under exact surrogate-control validity, $\delta_{\mathrm{SV}} = 0_p$, we have $\beta_{\mathrm{SV}} = \beta$ and hence $\widehat{\beta}_{\mathrm{SV}} \xrightarrow{\mathrm{p}} \beta$. Identification requires sufficiently rich variation in x_i to remain after partialling out the surrogate controls; see Appendix A.2 for further discussion of the residualization geometry.

2.4 Causal interpretation

Theorems 1 and 2 identify the statistical targets $\beta_{\mathrm{IV}} = \beta + Q_{\mathrm{IV}}^{-1}\delta_{\mathrm{IV}}$ and $\beta_{\mathrm{SV}} = \beta + Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}$. A causal interpretation as the structural coefficient β therefore requires the corresponding projected bias to vanish. For IV, consistency, exclusion, and exogeneity, together with the homogeneous linear response $y_i(x) = \alpha_i + \beta^\top x$, imply $\mathbb{E}(z_i\varsigma_i) = 0_q$, hence $\delta_{\mathrm{IV}} = 0_p$ and $\beta_{\mathrm{IV}} = \beta$ (Wooldridge, 2010); under heterogeneous effects, the linear IV coefficient need not represent a common causal effect and may instead have a local interpretation under additional conditions (Imbens and Angrist, 1994). For SV, causal validity requires the surrogate controls to capture the confounding-relevant component of x_i, so that $\mathbb{E}[(\Pi_w^{\perp}x_i)\varsigma_i] = 0_p$, implying $\beta_{\mathrm{SV}} = \beta$. Thus, IV uses instrument-selected variation whereas SV uses residualized variation after surrogate adjustment; when the corresponding causal assumptions fail, the estimators remain well defined but generally target β_{IV} and β_{SV} rather than β.

3. Combining instruments and surrogate controls

The IV and SV approaches developed above yield two projected moment restrictions for the same structural parameter β, motivating a common GMM formulation. We study the projected GMM estimator $\widehat{\beta}(M)$, establish complementary identification, and derive its first-order behavior under exact and $n^{-1/2}$-local misspecification. The two regimes lead respectively to variance-optimal and bias–variance-optimal weighting. We finally consider a one-parameter weighted least-squares path as an interpretable restricted GMM family.

3.1 Projected GMM formulation and complementary identification

Consider the linear structural model (2.2). Under exact projected validity, $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$ and $\mathbb{E}\{(\Pi_w^{\perp} x_i)\varsigma_i\} = 0_p$, equivalently $\delta_{\text{IV}} = \delta_{\text{SV}} = 0_p$, so both projected moment systems target the structural coefficient β.

Define the stacked projected regressor $h_i := \{(\Pi_z x_i)^\top, (\Pi_w^{\perp} x_i)^\top\}^\top \in \mathbb{R}^{2p}$ and its Jacobian $J := \mathbb{E}(h_i x_i^\top) = \{Q_{\text{IV}}^\top, Q_{\text{SV}}^\top\}^\top$. The joint population restriction is

$$\mathbb{E}\{h_i(y_i - x_i^\top \beta)\} = 0_{2p}. \tag{3.1}$$

Proposition 3 (Complementary identification)**.** Suppose the projected mo-

ments in (3.1) are valid. Then β is uniquely identified by the stacked IV–SV system if and only if

$$\operatorname{rank}(J) = p \quad \Longleftrightarrow \quad \ker(Q_{\mathrm{IV}}) \cap \ker(Q_{\mathrm{SV}}) = \{0_p\}. \tag{3.2}$$

Thus, neither projected block needs to have full rank individually, provided that the two blocks do not lose the same nonzero parameter direction.

Indeed, equality of the stacked moments at θ and β is equivalent to $J(\theta - \beta) = 0$, and $\ker(J) = \ker(Q_{\mathrm{IV}}) \cap \ker(Q_{\mathrm{SV}})$. In particular, $\operatorname{rank}(Q_{\mathrm{IV}}) = \operatorname{rank}(\Sigma_{zx}) \leq q$, so $q < p$ rules out IV-only identification of the full coefficient but not identification by the stacked system. The SV block has no analogous requirement $m \geq p$; its identifying content is determined by $\operatorname{rank}(Q_{\mathrm{SV}})$.

For the sample counterpart, define $\widehat{g}(\theta) := n^{-1}\{[X^\top P_Z(Y - X\theta)]^\top, [X^\top P_W^\perp(Y - X\theta)]^\top\}^\top$. For a fixed $M \in \mathbb{S}_{++}^{2p}$, the projected GMM estimator is

$$\widehat{\beta}(M) := \operatorname*{argmin}_{\theta \in \mathbb{R}^p} \widehat{g}(\theta)^\top M \widehat{g}(\theta). \tag{3.3}$$

Because $\widehat{g}(\theta)$ is affine in θ, its first-order condition is linear. Whenever the required inverse exists,

$$\widehat{\beta}(M) = \left\{ \begin{pmatrix} X^\top P_Z X \\ X^\top P_W^\perp X \end{pmatrix}^\top M \begin{pmatrix} X^\top P_Z X \\ X^\top P_W^\perp X \end{pmatrix} \right\}^{-1} \begin{pmatrix} X^\top P_Z X \\ X^\top P_W^\perp X \end{pmatrix}^\top M \begin{pmatrix} X^\top P_Z Y \\ X^\top P_W^\perp Y \end{pmatrix}. \tag{3.4}$$

The common factor n^{-1} cancels from the first-order condition.

The family (3.3) contains the two single-source projection estimators whenever the corresponding block identifies the full coefficient vector: the IV block yields $(X^\top P_Z X)^{-1} X^\top P_Z Y = \widehat{\beta}_{\mathrm{IV}}$, whereas the SV block yields $(X^\top P_W^\perp X)^{-1} X^\top P_W^\perp Y = \widehat{\beta}_{\mathrm{SV}}$. A projected block may therefore contribute identifying directions to the stacked system without identifying β on its own. When $q > p$, the original q-dimensional IV moment may additionally contain overidentifying information not retained by the projected p-dimensional moment; this distinction is revisited in Section 4.

3.2 Asymptotic theory under exact and local specification

The feasible moments entering $\widehat{\beta}(M)$ use estimated projection operators, so their first-order behavior is not generally represented by $h_i \varsigma_i$ alone. Following the notation introduced for the single-source procedures, we continue to write $\psi_{\mathrm{IV},i}$ and $\psi_{\mathrm{SV},i}$ for the corresponding projected first-order terms. Under exact projected validity, $\beta_{\mathrm{IV}} = \beta_{\mathrm{SV}} = \beta$, so these coincide with the single-source scores defined earlier and can be written as

$$\begin{aligned} \psi_{\mathrm{IV},i} &= (\Pi_z x_i)\varsigma_i + \{x_i - \Pi_z x_i\} z_i^\top \Sigma_z^{-1} \mathbb{E}(z_i \varsigma_i), \\ \psi_{\mathrm{SV},i} &= (\Pi_w^\perp x_i)\{\varsigma_i - w_i^\top \Sigma_w^{-1} \mathbb{E}(w_i \varsigma_i)\}. \end{aligned} \tag{3.5}$$

Under local misspecification, the single-source target-centered scores would instead be defined around their corresponding locally drifting statistical targets. For the combined analysis below, however, we expand around the structural coefficient β and, with a slight abuse of notation, retain $\psi_{\mathrm{IV},i}$ and $\psi_{\mathrm{SV},i}$ for the structural-β-centered first-order terms in (3.5). This convention allows the exact and locally misspecified regimes to be treated within the same notation; under local misspecification these scores need not be centered. Stack $\psi_i := (\psi_{\mathrm{IV},i}^\top, \psi_{\mathrm{SV},i}^\top)^\top$, let $\Sigma := \mathrm{Var}(\psi_i)$, and define $K_M := (J^\top M J)^{-1} J^\top M$ and $V(M) := K_M \Sigma K_M^\top$.

Assumption 3 (Combined GMM conditions)**.** Suppose the following conditions hold.

(i) **Moment restriction and relevance.** Suppose $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$, $\mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} = 0_p$, $\Sigma_z \in \mathbb{S}_{++}^q$, $\Sigma_w \in \mathbb{S}_{++}^m$, and $\mathrm{rank}(J) = p$. The weighting matrix $M \in \mathbb{S}_{++}^{2p}$ is fixed.

(ii) **Second moments and nondegeneracy.** Suppose $\mathbb{E}\|x_i z_i^\top\|_{\mathrm{op}}^2 < \infty$, $\mathbb{E}\|z_i\{\varsigma_i - z_i^\top \Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)\}\|_2^2 < \infty$, $\mathbb{E}\|x_i\varsigma_i\|_2^2 < \infty$, $\mathbb{E}\|x_i w_i^\top\|_{\mathrm{op}}^2 < \infty$, $\mathbb{E}\|w_i\{\varsigma_i - w_i^\top \Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\}\|_2^2 < \infty$, and $V(M) \in \mathbb{S}_{++}^p$.

Estimating the projection operators contributes to the first-order covariance through (3.5). Thus, unlike classical GMM with directly ob-

served moment functions, the relevant score is ψ_i, rather than simply $h_i\varsigma_i$. Assumption 3(ii) ensures that the score covariance is well defined, while $V(M) \in \mathbb{S}^p_{++}$ allows redundant moment directions but ensures nondegenerate inference. Positive definiteness of Σ is only required when Σ^{-1} is used for efficient weighting.

Theorem 4 (Identification, consistency, and asymptotic normality under exact specification)**.** Under Assumption 3(i), the following hold. *(i) Identification.* The structural coefficient β is the unique solution to the population combined projected moment condition. *(ii) Consistency.* $\widehat{\beta}(M) \xrightarrow{\mathrm{p}} \beta$. *(iii) Asymptotic normality.* If, in addition, Assumption 3(ii) holds, then $\sqrt{n}\{\widehat{\beta}(M) - \beta\} \xrightarrow{\mathrm{d}} \mathcal{N}\{0, V(M)\}$.

We next retain the same estimator but allow the projected moments to fail at the $n^{-1/2}$ scale. Following Armstrong and Kolesár (2021), suppose

$$\mathbb{E}(h_i\varsigma_i) = \frac{\tau}{\sqrt{n}} = \frac{1}{\sqrt{n}} \begin{pmatrix} \tau_z \\ \\ \tau_w \end{pmatrix}, \qquad \tau_z, \tau_w \in \mathbb{R}^p. \tag{3.6}$$

Thus, $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = \tau_z/\sqrt{n}$ and $\mathbb{E}\{(\Pi_w^{\perp} x_i)\varsigma_i\} = \tau_w/\sqrt{n}$.

For the expansion around the structural coefficient β, retain the same score definitions in (3.5). Since $\mathbb{E}[\{x_i - \Pi_z x_i\} z_i^{\top}] = 0_{p\times q}$ and $\mathbb{E}\{(\Pi_w^{\perp} x_i) w_i^{\top}\} = 0_{p\times m}$, the projection correction terms have mean zero. Hence $\mathbb{E}(\psi_{\mathrm{IV},i}) =$

$\tau_z/\sqrt{n}$, $\mathbb{E}(\psi_{\text{SV},i}) = \tau_w/\sqrt{n}$, and $\mathbb{E}(\psi_i) = \tau/\sqrt{n}$. Estimating the projection operators therefore affects the first-order covariance but not the local misspecification vector.

Along the local sequence, recall $\Sigma := \text{Var}(\psi_i)$ and suppose $\Sigma \to \Sigma_{miss}$. Let $\Sigma_{\text{IV,miss}}$ and $\Sigma_{\text{SV,miss}}$ denote the diagonal blocks of Σ, and define $V_{\text{miss}}(M) := K_M \Sigma_{miss} K_M^\top$. Thus, $K_M\tau$ and $V_{\text{miss}}(M)$ are the local asymptotic bias and variance associated with M.

Assumption 4 (Combined GMM conditions under local misspecification)**.** Suppose the following conditions hold.

(i) **Local moment restriction and relevance.** Suppose (3.6) holds for a fixed $\tau \in \mathbb{R}^{2p}$, $\Sigma_z \in \mathbb{S}^q_{++}$, $\Sigma_w \in \mathbb{S}^m_{++}$, and the fixed symmetric weighting matrix $M \succeq 0$ satisfies $J^\top M J \in \mathbb{S}^p_{++}$.

(ii) **Second-moment conditions.** Suppose, uniformly along the local sequence, $\sup_n \mathbb{E}\|x_i z_i^\top\|^2_{\text{op}} < \infty$, $\sup_n \mathbb{E}\|z_i\{\varsigma_i - z_i^\top \Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)\}\|_2^2 < \infty$, $\sup_n \mathbb{E}\|x_i\varsigma_i\|_2^2 < \infty$, $\sup_n \mathbb{E}\|x_i w_i^\top\|^2_{\text{op}} < \infty$, and $\sup_n \mathbb{E}\|w_i\{\varsigma_i - w_i^\top \Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\}\|_2^2 < \infty$.

(iii) **Lindeberg condition, covariance convergence, and estimator nondegeneracy.** For every $\epsilon > 0$, suppose $\mathbb{E}[\|\psi_i - \mathbb{E}(\psi_i)\|_2^2 \mathbf{1}\{\|\psi_i - \mathbb{E}(\psi_i)\|_2 > \epsilon\sqrt{n}\}] \to 0$. Suppose further that, for some $\Sigma \in \mathbb{S}^{2p}_+$,

$\|\Sigma - \Sigma_{miss}\|_{\text{op}} \to 0$, and $V_{\text{miss}}(M) = K_M \Sigma K_M^\top \in \mathbb{S}^p_{++}$.

Theorem 5 (Combined GMM under local misspecification)**.** Under Assumption 4 (i), the following hold. *(i) Local identification.* The population GMM criterion has the unique target $\beta_{M,n} = \beta + n^{-1/2} K_M \tau$, and hence $\beta_{M,n} \to \beta$. *(ii) Consistency.* $\widehat{\beta}(M) \xrightarrow{\text{p}} \beta$. *(iii) Asymptotic normality.* If, in addition, Assumption 4 (ii)–(iii) hold, then $\sqrt{n}\{\widehat{\beta}(M) - \beta\} \xrightarrow{\text{d}} \mathcal{N}\{K_M \tau, V_{\text{miss}}(M)\}$. The same conclusions apply to the IV-only and SV-only systems under the corresponding specializations, provided that their respective identification conditions hold.

The exact and locally misspecified regimes therefore differ only at first order through the mean shift. Under exact specification, M affects asymptotic variance only; under local misspecification, it affects both the local bias $K_M \tau$ and variance $V_{\text{miss}}(M)$. Setting $\tau = 0$ recovers the exact specification benchmark and motivates the different optimal weighting criteria studied next.

4. Optimal weighting across specification regimes

The preceding results show that the weighting matrix affects only first-order variance under exact specification, but affects both local bias and variance under local misspecification. We now choose the weight by minimizing

asymptotic squared-error risk. Since the first-order behavior of $\widehat{\beta}(M)$ depends on M only through its sensitivity K_M, it is convenient to formulate the optimization directly in terms of K_M.

4.1 Risk-optimal weighting

For a fixed local misspecification vector τ, define the asymptotic mean squared error under squared Euclidean loss by

$$\mathrm{AMSE}_\tau(M) := \lim_{n\to\infty} n\,\mathbb{E}_\tau\big\|\widehat{\beta}(M) - \beta\big\|_2^2, \tag{4.1}$$

whenever the limit exists, where $\mathbb{E}_\tau$ denotes expectation along the local sequence indexed by τ. By Theorem 5, together with uniform integrability of the squared root-n estimation errors, $\mathrm{AMSE}_\tau(M) = \|K_M\tau\|_2^2 + \mathrm{tr}(K_M\Sigma K_M^\top)$. Thus AMSE separates naturally into squared local bias and asymptotic variance.

To allow different degrees of misspecification in the two projected moment blocks, let $D := \mathrm{diag}(c_z I_p, c_w I_p)$, with $c_z, c_w \geq 0$, and consider $\mathcal{C}(D) := \{\tau = Du : \|u\|_2 \leq 1\}$. Here c_z and c_w bound the local misspecification allowed in the IV and SV channels, respectively.

Lemma 6 (Worst-case asymptotic risk)**.** Under the conditions of Theorem 5 and the uniform-integrability condition above, $\mathrm{AMSE}_\tau(M) = \|K_M\tau\|_2^2 +$

$\mathrm{tr}(K_M \Sigma K_M^\top)$ for every fixed $\tau \in \mathcal{C}(D)$. Consequently, for any K satisfying $KJ = I_p$, the worst-case AMSE is

$$R(K) := \sup_{\tau \in \mathcal{C}(D)} \mathrm{AMSE}_\tau(K) = \|KD\|_{\mathrm{op}}^2 + \mathrm{tr}(K\Sigma K^\top). \tag{4.2}$$

The risk-optimal sensitivity is therefore defined by the convex problem

$$K_{\mathrm{opt}} \in \operatorname*{argmin}_{K:\, KJ = I_p} \left\{ \|KD\|_{\mathrm{op}}^2 + \mathrm{tr}(K\Sigma K^\top) \right\}. \tag{4.3}$$

This is equivalent to optimizing over admissible GMM weights. Every GMM sensitivity satisfies $K_M J = I_p$. Conversely, for any K satisfying $KJ = I_p$, let $N_J \in \mathbb{R}^{2p \times p}$ have orthonormal columns spanning $\ker(J^\top)$. Then the full class of positive-definite weights inducing K is $\mathcal{M}(K) = \left\{ K^\top A K + N_J B N_J^\top : A, B \in \mathbb{S}_{++}^p \right\}$. Indeed, for any $M \in \mathcal{M}(K)$, $J^\top M J = A$ and $J^\top M = AK$, so that $(J^\top M J)^{-1} J^\top M = K$; conversely, every $M \in \mathbb{S}_{++}^{2p}$ inducing K admits this representation. Hence the risk-optimal weights are precisely $\mathcal{M}(K_{\mathrm{opt}})$, while the optimal sensitivity itself is the relevant first-order object.

Problem (4.3) is convex: the constraint is affine, the variance term is quadratic, and the squared operator norm is convex. If $\Sigma \in \mathbb{S}_{++}^{2p}$, the objective is strongly convex, so K_{opt} is unique.

Under exact specification, $c_z = c_w = 0$, so $D = 0$ and the bias term vanishes. Replacing Σ by the exact-specification covariance Σ, the problem

reduces to minimizing $\operatorname{tr}(K\Sigma K^\top)$ subject to $KJ = I_p$. When $\Sigma \in \mathbb{S}^{2p}_{++}$, the unique solution is

$$K_{\text{opt}} = (J^\top \Sigma^{-1} J)^{-1} J^\top \Sigma^{-1}.$$

The usual efficient-GMM weight $M^\star = \Sigma^{-1}$ is a canonical inducing weight (Hansen, 1982), while the full class of weights inducing the same efficient sensitivity is $\mathcal{M}(K_{\text{opt}})$. Thus exact specification yields the usual variance-optimal GMM solution, whereas under local misspecification the additional term $\|KD\|^2_{\text{op}}$ trades potential moment violation against precision, as in Armstrong and Kolesár (2021).

The IV-only and SV-only procedures are special cases of (4.3). If $Q_{\text{IV}}, Q_{\text{SV}} \in \mathbb{S}^p_{++}$, their sensitivities are $K^{(\text{IV})} = (Q_{\text{IV}}^{-1}, 0)$ and $K^{(\text{SV})} = (0, Q_{\text{SV}}^{-1})$. Under local misspecification, $R\{K^{(\text{IV})}\} = c_z^2 \|Q_{\text{IV}}^{-1}\|^2_{\text{op}} + \operatorname{tr}(Q_{\text{IV}}^{-1}\Sigma_{\text{IV,miss}} Q_{\text{IV}}^{-T})$, while $R\{K^{(\text{SV})}\} = c_w^2 \|Q_{\text{SV}}^{-1}\|^2_{\text{op}} + \operatorname{tr}(Q_{\text{SV}}^{-1}\Sigma_{\text{SV,miss}} Q_{\text{SV}}^{-T})$. Under exact specification, $c_z = c_w = 0$, so only the variance terms remain.

A feasible version of (4.3) replaces J and the score covariance by consistent first-step estimators; its construction and consistency are given in Appendix F.

4.2 Risk dominance across specification regimes

The preceding formulation makes the comparison with the single-source procedures immediate. Whenever the corresponding block identifies the full coefficient vector, the IV-only and SV-only sensitivities are feasible points of the combined optimization problem. The optimally combined procedure can therefore do no worse under the same risk criterion.

Proposition 7 (Risk dominance under exact specification)**.** Suppose *(i)* the projected moment restrictions hold, $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$ and $\mathbb{E}\{(\Pi_w^{\perp} x_i)\varsigma_i\} = 0_p$, with $\Sigma_z \in \mathbb{S}^q_{++}$, $\Sigma_w \in \mathbb{S}^m_{++}$, and $\operatorname{rank}(J) = p$; *(ii)* the second-moment conditions in Assumption 3(ii) hold; *(iii)* $Q_{\text{IV}}, Q_{\text{SV}} \in \mathbb{S}^p_{++}$; and *(iv)* $\Sigma \in \mathbb{S}^{2p}_{++}$. Then the efficient combined estimator satisfies

$$\operatorname{tr}\left\{(J^\top \Sigma^{-1} J)^{-1}\right\} \leq \min\left[\operatorname{tr}\left(Q_{\text{IV}}^{-1}\Sigma_{\text{IV}}Q_{\text{IV}}^{-T}\right), \operatorname{tr}\left(Q_{\text{SV}}^{-1}\Sigma_{\text{SV}}Q_{\text{SV}}^{-T}\right)\right].$$

Thus the optimally combined projected system has asymptotic squared-error risk no larger than either identified single-source estimator.

This is the usual efficient-GMM comparison expressed under squared Euclidean loss. Condition *(iii)* ensures that both single-source benchmarks identify the full coefficient vector, while condition *(iv)* ensures that the efficient combined weight is well defined. The result follows because $K^{(\text{IV})} = (Q_{\text{IV}}^{-1}, 0)$ and $K^{(\text{SV})} = (0, Q_{\text{SV}}^{-1})$ both belong to the feasible class $\{K : KJ =$

$I_p\}$.

Proposition 8 (Risk dominance under local misspecification)**.** Suppose (i) Assumption 4 holds, (ii) $Q_{\text{IV}} \in \mathbb{S}^p_{++}$, (iii) $Q_{\text{SV}} \in \mathbb{S}^p_{++}$, and (iv) $\Sigma \in \mathbb{S}^{2p}_{++}$. Let K_{opt} solve (4.3). Then

$$R(K_{\text{opt}}) \leq R\{K^{(\text{IV})}\}, \qquad R(K_{\text{opt}}) \leq R\{K^{(\text{SV})}\}.$$

Hence the optimally combined IV–SV estimator has worst-case local AMSE no larger than either identified single-source procedure.

The two propositions have the same geometric explanation. The combined optimization contains the two single-source procedures as feasible special cases, while also allowing the estimator to use both projected moment blocks. The single-source comparisons require Q_{IV} or Q_{SV} to be nonsingular, respectively. By contrast, the combined procedure remains well defined whenever $\text{rank}(J) = p$. Thus an individual projected block may be rank deficient and still contribute identifying or precision-improving directions to the stacked system, even though its own full-vector risk is not defined as a benchmark.

Remark 1 (Misspecification-robust confidence intervals)**.** The same misspecification set can be used for robust inference. For a prespecified nonzero $a \in \mathbb{R}^p$, the worst-case local asymptotic bias of $a^\top \widehat{\beta}$ under sensitivity K is

$\sup_{\tau \in \mathcal{C}(D)} |a^\top K\tau| = \|DK^\top a\|_2$. Let $\widehat{b}_a := \|D\widehat{K}^\top a\|_2$ and $\widehat{s}_a^2 := a^\top \widehat{K}\widehat{\Sigma}\widehat{K}^\top a$. Following Armstrong and Kolesár (2021), a $100(1-\alpha)\%$ misspecification-robust confidence interval is $a^\top\widehat{\beta} \pm \mathrm{cv}_\alpha(\widehat{b}_a/\widehat{s}_a)\,\widehat{s}_a/\sqrt{n}$, where $\mathrm{cv}_\alpha(t)$ is the $1-\alpha$ quantile of $|Z|$ for $Z \sim \mathcal{N}(t,1)$. When $c_z = c_w = 0$, the bias bound is zero and the interval reduces to the usual Wald interval.

4.3 Restricted weighting: a one-parameter WLS path

The general projected GMM estimator allows a $2p \times 2p$ weighting matrix and remains the benchmark procedure. For interpretation, however, it is useful to consider a simpler one-parameter path that moves continuously between the IV and SV projection criteria. The resulting weighted least-squares estimator is not a separate estimation framework; it is a restricted subfamily of $\widehat{\beta}(M)$.

This restriction is deliberately narrower than the general stacked-GMM theory. To retain both endpoints $\lambda = 0$ and $\lambda = 1$, we work in the regime where the projected IV and SV blocks are individually nonsingular. Consequently, the full path considered below does not cover complementary-rank settings such as $q < p$, in which Theorem 4 may still identify β through the stacked system.

Recall that $\widehat{\beta}_{\mathrm{IV}}$ and $\widehat{\beta}_{\mathrm{SV}}$ minimize $\|P_Z(Y - Xb)\|_2^2$ and $\|P_W^\perp(Y - Xb)\|_2^2$,

respectively. For $\lambda \in [0,1]$, define $\widehat{\beta}_\lambda := \operatorname{argmin}_b\{(1-\lambda)\|P_Z(Y-Xb)\|_2^2 + \lambda\|P_W^\perp(Y-Xb)\|_2^2\}$. Its first-order condition gives

$$\widehat{\beta}_\lambda = \{(1-\lambda)X^\top P_Z X + \lambda X^\top P_W^\perp X\}^{-1}\{(1-\lambda)X^\top P_Z Y + \lambda X^\top P_W^\perp Y\}. \tag{4.4}$$

In particular, $\widehat{\beta}_0 = \widehat{\beta}_{\text{IV}}$ and $\widehat{\beta}_1 = \widehat{\beta}_{\text{SV}}$.

The next result makes explicit that this path is contained in the projected GMM family.

Proposition 9 (Restricted-GMM interpretation)**.** Suppose that $n^{-1}X^\top P_Z X \in \mathbb{S}_{++}^p$ and $n^{-1}X^\top P_W^\perp X \in \mathbb{S}_{++}^p$. For $\lambda \in (0,1)$, define

$$M_\lambda := \begin{pmatrix} (1-\lambda)\big(n^{-1}X^\top P_Z X\big)^{-1} & 0 \\ 0 & \lambda\big(n^{-1}X^\top P_W^\perp X\big)^{-1} \end{pmatrix}.$$

Then $M_\lambda \in \mathbb{S}_{++}^{2p}$, and $\widehat{\beta}_\lambda = \widehat{\beta}(M_\lambda)$. Thus, for $\lambda \in (0,1)$, the WLS estimator is a one-parameter restricted subfamily of the joint GMM estimator. At $\lambda = 0$ and $\lambda = 1$, the corresponding positive-semidefinite block weights recover $\widehat{\beta}_{\text{IV}}$ and $\widehat{\beta}_{\text{SV}}$, respectively.

Proposition 9 clarifies the role of the WLS path. The unrestricted GMM estimator can use general weighting, including off-diagonal blocks that account for dependence between the IV and SV scores. By contrast, M_λ restricts attention to a one-dimensional block-diagonal family. The WLS

path is therefore useful for interpreting how the estimator moves between the two projected criteria, but it is not the general efficiency or local-risk benchmark.

Having established this restricted-GMM interpretation, we next consider how λ trades off the two projected criteria within this one-dimensional family. This differs from the preceding analysis, which studies the unrestricted projected GMM estimator through unconditional first-order asymptotic criteria. To obtain a transparent finite-sample characterization of the restricted path, we condition on the realized design matrices (X, Z, W). This treats the information matrices along the path as fixed and isolates the contribution of the structural disturbance to estimation risk. The conditioning is only an analytical device and does not impose conditional validity of either projected moment restriction.

When β is vector-valued, there is generally no scalar λ that minimizes the variance matrix in the Loewner order. A scalar optimum therefore requires a scalar loss criterion. We use conditional trace risk.

For this analysis, define $S_Z := X^\top P_Z X$, $S_W := X^\top P_W^\perp X$, and $S_\lambda := (1-\lambda)S_Z + \lambda S_W$. Let $\mathcal{D} := \sigma(X, Z, W)$, $U_Z := X^\top P_Z \varsigma$, $U_W := X^\top P_W^\perp \varsigma$, $c_Z := \mathbb{E}(\|U_Z\|_2^2 \mid \mathcal{D})$, $c_{ZW} := \mathbb{E}(U_Z^\top U_W \mid \mathcal{D})$, and $c_W := \mathbb{E}(\|U_W\|_2^2 \mid \mathcal{D})$.

Proposition 10 (Conditional trace risk and a tractable surrogate)**.** Work

conditionally on the realized design matrices (X, Z, W), treated as fixed throughout. Suppose the following conditions hold.

(i) $S_Z \in \mathbb{S}^p_{++}$ and $S_W \in \mathbb{S}^p_{++}$, or equivalently $S_\lambda \in \mathbb{S}^p_{++}$ for every $\lambda \in [0,1]$.

(ii) $c_Z < \infty$ and $c_W < \infty$.

Then $\widehat{\beta}_\lambda - \beta = S_\lambda^{-1}\{(1-\lambda)U_Z + \lambda U_W\}$. Let $U_\lambda := (1-\lambda)U_Z + \lambda U_W$. The conditional trace risk is

$$\begin{aligned}\mathcal{R}_{\mathrm{tr}}(\lambda) &:= \mathbb{E}(\|\widehat{\beta}_\lambda - \beta\|_2^2 \mid \mathcal{D}) \\ &= \mathrm{tr}\left[S_\lambda^{-1}\mathbb{E}(U_\lambda U_\lambda^\top \mid \mathcal{D})S_\lambda^{-1}\right]. \end{aligned} \tag{4.5}$$

Equivalently, $\mathcal{R}_{\mathrm{tr}}(\lambda) = \|S_\lambda^{-1}\mathbb{E}(U_\lambda \mid \mathcal{D})\|_2^2 + \mathrm{tr}\{S_\lambda^{-1}\,\mathrm{Var}(U_\lambda \mid \mathcal{D})S_\lambda^{-1}\}$.

Moreover, $\mathrm{tr}\{\mathbb{E}(U_\lambda U_\lambda^\top \mid \mathcal{D})\} = (1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W$, and therefore

$$\begin{aligned}\frac{(1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W}{\lambda_{\max}(S_\lambda)^2} &\le \mathcal{R}_{\mathrm{tr}}(\lambda) \\ &\le \frac{(1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W}{\lambda_{\min}(S_\lambda)^2}.\end{aligned} \tag{4.6}$$

If, in addition, $\lambda_{\max}(S_Z) \lesssim \lambda_{\min}(S_Z)$ and $\lambda_{\max}(S_W) \lesssim \lambda_{\min}(S_W)$, then uniformly over $\lambda \in [0,1]$,

$$\mathcal{R}_{\mathrm{tr}}(\lambda) \asymp \widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda)$$

$$:= \frac{(1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W}{\{(1-\lambda)\lambda_{\min}(S_Z) + \lambda\lambda_{\min}(S_W)\}^2}. \tag{4.7}$$

If, moreover, $(c_W - c_{ZW})\lambda_{\min}(S_Z) + (c_Z - c_{ZW})\lambda_{\min}(S_W) \neq 0$, then $\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda)$ has at most one interior stationary point, given by

$$\lambda^\star_{\mathrm{sur}} = \frac{c_Z\lambda_{\min}(S_W) - c_{ZW}\lambda_{\min}(S_Z)}{(c_W - c_{ZW})\lambda_{\min}(S_Z) + (c_Z - c_{ZW})\lambda_{\min}(S_W)}. \tag{4.8}$$

Consequently, the minimizer of $\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda)$ over $[0,1]$ is obtained by comparing its values at $\lambda = 0$, $\lambda = 1$, and $\lambda = \lambda^\star_{\mathrm{sur}}$ whenever $\lambda^\star_{\mathrm{sur}} \in (0,1)$.

Proposition 10 is a finite-sample conditional risk characterization and does not require the projected score blocks to be conditionally centered. In particular, the unconditional IV and SV restrictions do not generally imply $\mathbb{E}(U_Z \mid \mathcal{D}) = 0$ or $\mathbb{E}(U_W \mid \mathcal{D}) = 0$. Indeed,

$$\begin{aligned}
c_Z &= \mathrm{tr}\{X^\top P_Z \,\mathrm{Var}(\varsigma \mid \mathcal{D}) P_Z X\} + \|\mathbb{E}(U_Z \mid \mathcal{D})\|_2^2, \\
c_{ZW} &= \mathrm{tr}\{X^\top P_Z \,\mathrm{Var}(\varsigma \mid \mathcal{D}) P_W^\perp X\} + \mathbb{E}(U_Z \mid \mathcal{D})^\top \mathbb{E}(U_W \mid \mathcal{D}), \\
c_W &= \mathrm{tr}\{X^\top P_W^\perp \,\mathrm{Var}(\varsigma \mid \mathcal{D}) P_W^\perp X\} + \|\mathbb{E}(U_W \mid \mathcal{D})\|_2^2.
\end{aligned}$$

Thus, c_Z, c_{ZW}, and c_W measure conditional second moments and incorporate both conditional bias and conditional variance. Under conditional centering, they reduce to covariance-based expressions.

The information matrix S_λ changes linearly with λ, whereas the conditional score second moment changes quadratically through c_Z, c_{ZW}, and

c_W. The bounds in (4.6) separate these two components and motivate the tractable surrogate criterion in (4.7). The candidate $\lambda^\star_{\text{sur}}$ therefore provides an interpretable scalar balance between the information strength and total conditional second moments of the two projected channels.

The unrestricted projected GMM estimator remains the benchmark procedure. Under exact specification its optimal weight is $M^\star = \Sigma^{-1}$, whereas under local misspecification its weight is chosen according to (4.3). The WLS path instead provides a transparent one-dimensional representation of the same IV–SV combination problem.

5. Simulation

We use two scalar experiments to illustrate the finite-sample implications of the preceding weighting results. The first considers exact specification and examines the efficiency gain from combining the IV and SV projected moments. The second introduces joint $n^{-1/2}$-local misspecification and evaluates the corresponding worst-case bias–variance tradeoff. Throughout, all projection operators and data-dependent GMM weights are re-estimated within each replication, and each design is based on 1,000 Monte Carlo replications.

5.1 Efficiency under correct specification

Consider the scalar design

$$z_i = s_i + \eta_i, \qquad w_i = (u_i, s_i)^\top, \qquad x_i = u_i + a s_i + v_i, \qquad y_i = \beta x_i + u_i + \varepsilon_i, \tag{5.1}$$

where $s_i, \eta_i, u_i, v_i, \varepsilon_i$ are mutually independent standard normal variables. We set $a = 2$, which gives a balanced design in which the IV-only and SV-only estimators both have asymptotic variance 1. The equal-weight and efficient combined GMM estimators have asymptotic variances 0.68 and 0.50, respectively. Thus the design provides a direct finite-sample illustration of the efficiency gain from combining the two projected moment blocks.

For point estimation, we set $\beta = 1$ and report n times empirical mean squared error. For inference, we consider two-sided Wald tests at the 5% level, evaluating Type I error under $H_0 : \beta = 0$ and local power under $\beta = 2/\sqrt{n}$.

The results in Table 1 closely track the population benchmarks. The scaled MSE of efficient GMM remains near its asymptotic value 0.50 and is uniformly below those of the two single-source estimators. Equal-weight GMM also improves substantially on either single-source procedure, but does not attain the efficient benchmark. The Wald tests are approximately

Table 1: Finite-sample performance under correct specification, $a = 2$.

Panel A. Scaled MSE

n	IV	SV	Equal	Efficient
200	1.079	1.044	0.690	0.532
400	1.099	1.032	0.730	0.550
600	0.972	1.033	0.646	0.493
800	1.020	1.030	0.695	0.529
1000	0.968	0.951	0.666	0.498
1200	1.095	1.036	0.726	0.512
1400	1.091	1.015	0.729	0.528
1600	0.984	1.030	0.667	0.508
1800	0.918	1.085	0.632	0.513
2000	1.047	0.947	0.697	0.486
Pop.	1.000	1.000	0.680	0.500

Panel B. Type I error

n	IV	SV	Equal	Efficient
200	0.050	0.058	0.047	0.052
400	0.058	0.047	0.062	0.061
600	0.051	0.062	0.050	0.062
800	0.045	0.062	0.049	0.061
1000	0.042	0.064	0.046	0.057
1200	0.052	0.057	0.050	0.058
1400	0.055	0.038	0.053	0.054
1600	0.052	0.051	0.058	0.057
1800	0.049	0.050	0.049	0.049
2000	0.064	0.038	0.061	0.041

Panel C. Local power

n	IV	SV	Equal	Efficient
200	0.516	0.530	0.676	0.805
400	0.517	0.505	0.694	0.819
600	0.528	0.509	0.680	0.806
800	0.538	0.511	0.671	0.800
1000	0.497	0.503	0.686	0.831
1200	0.522	0.544	0.672	0.812
1400	0.520	0.524	0.684	0.811
1600	0.544	0.529	0.697	0.829
1800	0.499	0.525	0.686	0.806
2000	0.498	0.528	0.667	0.809

Note: Panel A reports n times empirical MSE at $\beta = 1$; the Population row gives the corresponding asymptotic variances. Panel B reports rejection rates for two-sided 5% Wald tests of $H_0 : \beta = 0$. Panel C reports rejection rates under the local alternative $\beta = 2/\sqrt{n}$. "IV," "SV," "Equal," and "Efficient" denote the IV-only, SV-only, equal-weight GMM, and efficient GMM estimators, respectively.

size controlled across all four estimators. The reduction in variance translates directly into higher local power: at $n = 2000$, the rejection probability is 0.809 for efficient GMM, compared with 0.667 for equal-weight GMM, 0.498 for IV only, and 0.528 for SV only.

5.2 Risk under joint local misspecification

We examine the local bias–variance tradeoff in the balanced scalar design. Both projected moment blocks may be locally misspecified, with

$$\mathcal{C}(D) = \{(\tau_z, \tau_w)^\top : \tau_z^2 + \tau_w^2 \leq 1\}.$$

Thus, the joint misspecification radius is $\kappa = 1$. We evaluate 16 equally spaced points on the boundary of this set, using 1,000 replications at each sample size. The methods are IV only, SV only, equal-weight GMM, and feasible local-risk GMM.

Panel A reports the maximum of n times empirical MSE across the evaluated boundary points. Panels B and C use the same test obtained by inverting each method's 95% misspecification-robust confidence interval. Panel B reports its maximum Type I error under $H_0 : \beta = 1$; Panel C reports its minimum local power under $\beta = 1 + 2/\sqrt{n}$.

The local-risk estimator has the lowest worst-case scaled MSE at every reported sample size. At $n = 2000$, its value is 0.786, compared with 0.904

Table 2: Finite-sample performance under joint local misspecification, $\kappa = 1$.

Panel A. Worst-case scaled MSE

n	IV	SV	Equal	Local
200	1.405	1.980	0.907	0.785
400	1.321	2.090	0.897	0.793
600	1.261	2.062	0.903	0.801
800	1.318	2.110	0.915	0.820
1000	1.326	2.165	0.905	0.802
1200	1.284	2.048	0.901	0.786
1400	1.292	2.003	0.880	0.759
1600	1.239	2.075	0.881	0.795
1800	1.350	2.032	0.918	0.788
2000	1.296	1.990	0.904	0.786
Pop.	1.250	2.000	0.880	0.769

Panel B. Robust Type I error

n	IV	SV	Equal	Local
200	0.067	0.050	0.064	0.059
400	0.072	0.051	0.073	0.074
600	0.058	0.060	0.066	0.070
800	0.053	0.056	0.048	0.049
1000	0.055	0.065	0.053	0.052
1200	0.061	0.052	0.065	0.055
1400	0.061	0.050	0.059	0.055
1600	0.043	0.059	0.047	0.052
1800	0.055	0.050	0.055	0.062
2000	0.058	0.051	0.056	0.054

Panel C. Minimum robust local power

n	IV	SV	Equal	Local
200	0.274	0.050	0.395	0.419
400	0.296	0.051	0.388	0.408
600	0.288	0.049	0.405	0.419
800	0.254	0.039	0.332	0.363
1000	0.254	0.055	0.352	0.391
1200	0.269	0.049	0.388	0.417
1400	0.256	0.050	0.364	0.404
1600	0.273	0.045	0.381	0.396
1800	0.261	0.050	0.362	0.407
2000	0.260	0.048	0.356	0.376

Note: All extrema are taken over the 16 evaluated boundary points. Panel A uses $\beta = 1$; the Population row gives theoretical worst-case asymptotic risks. Panel B gives the maximum rejection rate of $H_0 : \beta = 1$, obtained by inverting each method's 95% misspecification-robust confidence interval. Equivalently, each entry is one minus that method's minimum robust coverage. Panel C gives the minimum rejection rate of the same robust test under $\beta = 1 + 2/\sqrt{n}$. "Local" denotes the local-risk GMM estimator.

for equal-weight GMM; its theoretical worst-case risk is 0.769. The robust Type I error rates in Panel B are generally close to the nominal 0.05: at $n = 2000$, they are 0.058, 0.051, 0.056, and 0.054 for IV, SV, equal-weight, and local-risk GMM, respectively. With the same robust test, local-risk GMM has the highest minimum local power at every reported sample size; at $n = 2000$, its power is 0.376, compared with 0.356 for equal-weight GMM.

6. Real data

In the single-cell CRISPR application, let $Z \in \mathbb{R}^{n\times q}$ denote the one-hot perturbation-condition matrix, $X \in \mathbb{R}^{n\times p}$ the expression matrix for target genes, $Y \in \mathbb{R}^{n\times g}$ the expression matrix for outcome genes, and $W \in \mathbb{R}^{n\times m}$ the surrogate-control matrix constructed from negative-control or non-differentially expressed genes. Although this notation accommodates a joint multi-perturbation model $Y \sim X$, our primary analysis considers one perturbation target at a time. For target k, we use the scalar instrument $Z_{\cdot k}$, the target expression $X_{\cdot k}$, and the multivariate outcome matrix Y. Thus, the real-data analysis reflects the common regime in which each target has a low-dimensional perturbation instrument, while the surrogate-control matrix W provides richer information about latent cell-state and technical variation.

7. Discussion

- nonlinear models: approximated by linear
- GMM with measurement error in variables Evdokimov and Zeleneev (2025)

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Department of Statistics and Actuarial Science, The University of Hong Kong

E-mail: haoranlei@connect.hku.hk

Department of Statistics and Actuarial Science and Musketeers Foundation Institute of Data Science, The University of Hong Kong

E-mail: jinhongd@hku.hk

A. Additional discussion on single-source deconfounding and causal interpretation

A.1 Additional details on the projected IV formulation

This subsection provides additional discussion of the projected IV formulation used in Section 2.2, including its relation to the classical IV moment, an explicit first-stage representation, and the simplification of the first-order theory under valid instruments.

Dimension and identification of the projected IV moment. Recall that $Q_{\text{IV}} = \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx}$. When $\Sigma_z \in \mathbb{S}^q_{++}$, $\operatorname{rank}(Q_{\text{IV}}) = \operatorname{rank}(\Sigma_{zx}) \le \min(p, q)$. Hence, identification of the full p-dimensional coefficient vector from the IV block alone requires $\operatorname{rank}(Q_{\text{IV}}) = p$, and therefore necessarily $q \ge p$.

When $q < p$, the projected IV moment $\mathbb{E}[(\Pi_z x_i)(y_i - x_i^\top \theta)]$ is still written as a p-dimensional vector, but its Jacobian has rank at most $q < p$. Thus, the p displayed equations contain fewer than p independent identifying directions and cannot uniquely identify the full coefficient vector.

The distinction between the raw and projected IV formulations is also relevant when $q \ge p$. The classical IV restriction $\mathbb{E}[z_i(y_i - x_i^\top \theta)] = 0_q$ contains q moment equations and, under $\operatorname{rank}(\Sigma_{zx}) = p$, is just identified when $q = p$ and overidentified when $q > p$. By contrast, $\mathbb{E}[(\Pi_z x_i)(y_i - x_i^\top \theta)] = \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}[z_i(y_i - x_i^\top \theta)]$ is always p-dimensional. Under valid IV, $\mathbb{E}(z_i \varsigma_i) = 0_q$, the raw and projected moments identify the same structural coefficient under the corresponding rank condition. When $q > p$, however, projection may discard overidentifying di-

rections contained in the raw IV moment. Consequently, efficiency statements for the projected IV system are not, in general, efficiency statements for GMM based on all raw IV moments.

Relation to an explicit first-stage representation. Some formulations of instrumental variables additionally posit $x_i = \eta^\top z_i + \epsilon_i$ with $\mathbb{E}(z_i \epsilon_i^\top) = 0$. Such a representation is useful when a structural interpretation of the first-stage regression is desired, but it is not required for the projected-moment formulation used here.

Indeed, whenever $\Sigma_z \in \mathbb{S}^q_{++}$, the population linear projection is defined directly as $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i$. Thus, the projected-IV statistical target and the corresponding 2SLS estimating equation depend only on this projection, the projected moment bias $\delta_{\text{IV}} := \mathbb{E}[(\Pi_z x_i)\varsigma_i]$, and the relevance matrix $Q_{\text{IV}} := \mathbb{E}[(\Pi_z x_i)x_i^\top]$. Accordingly, an explicit decomposition $x_i = \eta^\top z_i + \epsilon_i$ may be viewed as an optional linear-projection representation rather than a primitive assumption.

First-order representation under valid IV. Under the general projected-IV target, the first-order term is $\psi_{\text{IV},i} = (\Pi_z x_i)e_{\text{IV},i} + x_i z_i^\top \Sigma_z^{-1}\mathbb{E}[z_i(y_i - x_i^\top \beta_{\text{IV}})]$. The condition $\mathbb{E}[\|x_i z_i^\top\|_{\text{op}}^2] < \infty$ controls the second moment of the second component, whereas $\mathbb{E}\|z_i e_{\text{IV},i}\|_2^2 < \infty$ controls the projected-score component because $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i$.

Under valid IV, $\mathbb{E}(z_i\varsigma_i) = 0_q$, we have $\delta_{\text{IV}} = 0_p$, $\beta_{\text{IV}} = \beta$, and $\mathbb{E}[z_i(y_i - x_i^\top \beta_{\text{IV}})] = 0_q$. Hence, the correction term vanishes and $e_{\text{IV},i} = \varsigma_i$, so that $\psi_{\text{IV},i} = (\Pi_z x_i)\varsigma_i$. Therefore, $\mathbb{E}[\|x_i z_i^\top\|_{\text{op}}^2] < \infty$ is unnecessary for asymptotic normality in the valid-IV case; it suffices to impose the corresponding finite-second-moment condition on $z_i\varsigma_i$, together with nondegeneracy of the limiting

covariance matrix.

A.2 Additional details on surrogate-control residualization

This subsection provides further interpretation of the surrogate-control formulation in Section 2.3, with emphasis on the geometry of residualization, its projection representation, and the simplification of the first-order theory under exact surrogate-control validity.

Residualized relevance and projection geometry. Recall that $\Pi_w^{\perp} x_i = x_i - \Sigma_{xw}\Sigma_w^{-1} w_i$ and $Q_{\text{SV}} = \mathbb{E}[(\Pi_w^{\perp} x_i) x_i^{\top}]$. Since the population residual $\Pi_w^{\perp} x_i$ is orthogonal to w_i, we have

$$Q_{\text{SV}} = \mathbb{E}[(\Pi_w^{\perp} x_i)(\Pi_w^{\perp} x_i)^{\top}].$$

Hence, $Q_{\text{SV}} \in \mathbb{S}_{++}^{p}$ requires the residualized regressor to retain variation in every direction of the p-dimensional coefficient space.

This differs geometrically from IV relevance. The IV projection retains the component of x_i explained by z_i, whereas surrogate-control residualization removes the component of x_i explained by w_i. Thus, IV identification requires the projected component to span enough directions, whereas SV identification requires sufficiently rich variation to remain after partialling out the surrogate controls. This distinction motivates combining the two sources in the stacked moment system developed in the main text.

Partialling out as a projection representation. The surrogate-control estimator does not require a structural equation describing how w_i generates x_i. Its population construction only requires the linear projection $\Pi_w x_i = \Sigma_{xw}\Sigma_w^{-1} w_i$ and the residual $\Pi_w^{\perp} x_i = x_i - \Pi_w x_i$.

Equivalently, writing $\Gamma_w := \Sigma_w^{-1}\Sigma_{wx}$, one has the population decomposition $x_i = \Gamma_w^{\top} w_i + \Pi_w^{\perp} x_i$, with $\mathbb{E}[(\Pi_w^{\perp} x_i) w_i^{\top}] = 0$. The statistical target is determined by the residualized moment $\mathbb{E}[(\Pi_w^{\perp} x_i)(y_i - x_i^{\top}\theta)]$, rather than by a structural model for the relationship between x_i and w_i.

In particular, w_i itself need not be orthogonal to the structural disturbance. Under exact surrogate-control validity, $\mathbb{E}[(\Pi_w^{\perp} x_i)\varsigma_i] = 0_p$, which is equivalent to

$$\mathbb{E}(x_i \varsigma_i) = \Sigma_{xw}\Sigma_w^{-1}\mathbb{E}(w_i \varsigma_i).$$

Thus, the surrogate controls absorb the component of regressor–disturbance dependence represented linearly through w_i, leaving residualized variation orthogonal to the disturbance.

First-order representation under exact surrogate-control validity. Under the general statistical target $\beta_{\mathrm{SV}} = \beta + Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}$, define $r_{\mathrm{SV},i} := y_i - x_i^{\top}\beta_{\mathrm{SV}}$ and

$$e_{\mathrm{SV},i} := r_{\mathrm{SV},i} - w_i^{\top}\Sigma_w^{-1}\mathbb{E}(w_i r_{\mathrm{SV},i}).$$

Then $\mathbb{E}(w_i e_{\mathrm{SV},i}) = 0_m$, and the first-order term is $\psi_{\mathrm{SV},i} = (\Pi_w^{\perp} x_i) e_{\mathrm{SV},i}$.

The simplification relative to the IV case follows from the exact annihilation property $P_W^{\perp} W = 0$. After recentering $r_{\mathrm{SV},i}$ with respect to w_i, the component in the surrogate-control

span is removed by the sample residual-maker. By contrast, $P_Z Z = Z$, so the corresponding instrument-span component is retained in the IV case and generates an additional first-order correction term under projected moment bias.

Under exact surrogate-control validity, $\delta_{\text{SV}} = 0_p$, we have $\beta_{\text{SV}} = \beta$ and $r_{\text{SV},i} = \varsigma_i$, so

$$e_{\text{SV},i} = \varsigma_i - w_i^\top \Sigma_w^{-1} \mathbb{E}(w_i \varsigma_i).$$

Exact SV validity does not generally imply $\mathbb{E}(w_i \varsigma_i) = 0_m$, and hence $e_{\text{SV},i}$ need not equal ς_i. The first-order term therefore remains $\psi_{\text{SV},i} = (\Pi_w^\perp x_i) e_{\text{SV},i}$.

However, because $r_{\text{SV},i} = \varsigma_i$, the condition $\mathbb{E}\|x_i x_i^\top\|_{\text{op}}^2 < \infty$, which is needed under the general target to control the contribution of $x_i^\top Q_{\text{SV}}^{-1} \delta_{\text{SV}}$, is unnecessary for asymptotic normality under exact surrogate-control validity. The remaining second-moment conditions continue to control $x_i \varsigma_i$, the interaction between x_i and w_i, and the residualized score.

A.3 Additional discussion on causal interpretation

The results in Sections 2.2 and 2.3 are stated in terms of statistical moment conditions and statistical targets. This subsection provides additional discussion of the structural assumptions under which the no-bias conditions $\delta_{\text{IV}} = 0$ and $\delta_{\text{SV}} = 0$ admit causal interpretations.

IV moments and causal direction. In the linear IV formulation, exogeneity is commonly represented by $\mathbb{E}(z_i \varsigma_i) = 0_q$, while the projected formulation used in this paper requires the corresponding projected no-bias condition $\delta_{\text{IV}} := \mathbb{E}[(\Pi_z x_i) \varsigma_i] = 0_p$. These moment

restrictions are statistical statements and do not, by themselves, encode a causal direction between z_i and x_i.

The causal structure matters because it determines whether such orthogonality is scientifically plausible. For example, the classical IV interpretation views z_i as generating variation in x_i while remaining unrelated to the latent determinants of the outcome. By contrast, if z_i is constructed from, or strongly driven by, the endogenous regressor in a data-generating mechanism resembling $\varsigma_i \to x_i \to z_i$, then z_i may inherit information about ς_i, making the no-bias restriction difficult to justify. In such cases, the projected estimator remains statistically defined, but its target is $\beta_{\text{IV}} = \beta + Q_{\text{IV}}^{-1}\delta_{\text{IV}}$.

Heterogeneous causal effects under IV. A common structural interpretation of β additionally relies on a homogeneous linear response. If the potential outcome instead takes the form $y_i(x) = \alpha_i + \beta_i^\top x$, or is nonlinear in x, the IV coefficient generally need not represent a universal structural slope.

For binary treatment, under the usual additional conditions including monotonicity, IV identifies a local average treatment effect for compliers rather than a population-wide common effect (Imbens and Angrist, 1994). More generally, linear IV should be interpreted as summarizing causal responses associated with the variation in x_i induced by the instrument, rather than automatically as the entire causal response surface.

This distinction is separate from moment invalidity. Even when $\delta_{\text{IV}} = 0$, effect heterogeneity may alter the causal interpretation of the IV estimand; when $\delta_{\text{IV}} \neq 0$, the target additionally

contains the shift $Q_{\text{IV}}^{-1}\delta_{\text{IV}}$ induced by violation of the projected moment restriction.

Surrogate controls and causal direction. For surrogate-control adjustment, the statistical no-bias condition is $\delta_{\text{SV}} := \mathbb{E}[(\Pi_w^{\perp} x_i)\varsigma_i] = 0_p$. A natural causal justification is that w_i is a pre-treatment proxy for the latent confounding structure and captures the component of x_i associated with the structural disturbance.

The temporal or causal role of w_i is therefore important. If the data-generating mechanism is instead closer to $\varsigma_i \to x_i \to w_i$, then w_i may contain variation generated by the endogenous regressor itself. Partialling out such a variable need not remove only latent confounding and may invalidate the desired causal interpretation. The estimator nevertheless remains centered at the statistical target $\beta_{\text{SV}} = \beta + Q_{\text{SV}}^{-1}\delta_{\text{SV}}$, with $Q_{\text{SV}}^{-1}\delta_{\text{SV}}$ representing the remaining departure from the structural coefficient.

Relation to auxiliary-variable and proximal approaches. The surrogate-control formulation considered here is related to auxiliary-variable and proxy-based approaches to latent confounding. In particular, the auxiliary-variable framework of Miao et al. (2023) accommodates variables satisfying suitable exclusion-type conditions and develops nonparametric identification through equivalence and completeness conditions. At a high level, both approaches use observed ancillary information without requiring direct recovery of the latent confounder.

The identification targets and mechanisms are nevertheless different. The framework of

A.3 Additional discussion on causal interpretation

Miao et al. (2023) concerns nonparametric identification of potential-outcome distributions, whereas the present paper focuses on a linear structural coefficient and the projection, rank, first-order, and weighting structure generated by IV and surrogate-control moments.

The present formulation is also related in spirit to proximal causal inference (Miao et al., 2018, 2024). Proximal methods typically introduce treatment-inducing and outcome-inducing proxies and identify causal effects through conditional moment equations involving a bridge function. Their identification theory relies on conditions such as proxy validity, existence of an appropriate bridge function, and completeness.

Our framework instead uses the direct residualized moment $\mathbb{E}[(\Pi_w^{\perp} x_i)(y_i - x_i^{\top}\theta)] = 0_p$. We therefore do not solve a nonparametric bridge equation or attempt to recover the latent confounder. The surrogate variable enters through linear projection geometry: the component of x_i explained by w_i is removed, and inference uses the remaining variation.

Accordingly, these approaches should be viewed as complementary rather than equivalent. Auxiliary-variable and proximal methods provide broader nonparametric causal identification theories under proxy-variable assumptions, whereas the present formulation imposes a linear working structure to obtain a transparent projected-moment representation that is directly comparable with the IV channel and can subsequently be combined with it within the unified GMM framework.

B. Notation

Population L_2 projections are denoted by Π_z, Π_w, and $\Pi_w^{\perp}$, whereas for any sample matrix A with n rows, $\mathcal{C}(A)$ denotes its column space, P_A the Euclidean projection onto $\mathcal{C}(A)$, and $P_A^{\perp} := I_n - P_A$ the corresponding residual-maker. For any matrix A, $\ker(A) := \{v : Av = 0\}$ denotes its null space and $\operatorname{rank}(A) := \dim \mathcal{C}(A)$ its rank.

We use $\|\cdot\|_2$, $\|\cdot\|_{\mathrm{F}}$, and $\|\cdot\|_{\mathrm{op}}$ for the Euclidean, Frobenius, and operator norms, respectively, and $\operatorname{Cov}(a, b)$ for population cross-covariance.

We write $\xrightarrow{\mathrm{p}}$ and $\xrightarrow{\mathrm{d}}$ for convergence in probability and distribution, use $O_p(\cdot)$, $o_p(\cdot)$, and $\Theta_p(\cdot)$ in their standard senses, and write $a_n \lesssim b_n$ and $a_n \asymp b_n$ for one-sided and two-sided order equivalence up to constants.

The sets of $d \times d$ positive definite and positive semidefinite matrices are denoted by $\mathbb{S}_{++}^d$ and $\mathbb{S}_{+}^d$, respectively.

We use lower-case letters for unit-level variables and upper-case letters for sample arrays: $y_i \in \mathbb{R}$, $x_i \in \mathbb{R}^p$, $z_i \in \mathbb{R}^q$, $w_i \in \mathbb{R}^m$, and $u_i \in \mathbb{R}^r$ denote the outcome, regressors, instruments, surrogate controls, and latent confounders, respectively, with Y, X, Z, W, U denoting their corresponding sample arrays.

C. Proof

C.1 Proof of Theorem 1

Proof of Theorem 1. Recall that $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i$, $Q_{\text{IV}} = \mathbb{E}[(\Pi_z x_i)x_i^\top] = \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx}$, $\delta_{\text{IV}} = \mathbb{E}[(\Pi_z x_i)\varsigma_i]$, and $\beta_{\text{IV}} = \beta + Q_{\text{IV}}^{-1}\delta_{\text{IV}}$.

(i) Identification. Recall that the projected IV population moment condition is $\mathbb{E}[(\Pi_z x_i)(y_i - x_i^\top\theta)] = 0_p$, where $\theta \in \mathbb{R}^p$ is a candidate coefficient vector. Using $y_i = x_i^\top\beta + \varsigma_i$, we have

$$\mathbb{E}\Big[(\Pi_z x_i)\big(y_i - x_i^\top\theta\big)\Big] = \mathbb{E}\Big[(\Pi_z x_i)x_i^\top\Big](\beta - \theta) + \mathbb{E}[(\Pi_z x_i)\varsigma_i]\,.$$

By the definitions $Q_{\text{IV}} := \mathbb{E}[(\Pi_z x_i)x_i^\top]$ and $\delta_{\text{IV}} := \mathbb{E}[(\Pi_z x_i)\varsigma_i]$, the population moment condition is therefore equivalent to

$$Q_{\text{IV}}(\theta - \beta) = \delta_{\text{IV}}. \tag{C.1}$$

By Assumption 1(i), $Q_{\text{IV}} \in \mathbb{S}_{++}^p$, and hence Q_{IV} is nonsingular. Left-multiplying (C.1) by Q_{IV}^{-1} gives

$$\theta = \beta + Q_{\text{IV}}^{-1}\delta_{\text{IV}} = \beta_{\text{IV}}.$$

Thus a solution exists.

To see that the solution is unique, suppose that both θ_1 and θ_2 satisfy (C.1). Then $Q_{\text{IV}}(\theta_1 - \theta_2) = 0_p$. Since Q_{IV} is nonsingular, its null space contains only 0_p, so $\theta_1 - \theta_2 = 0_p$, and hence $\theta_1 = \theta_2$. Therefore, the projected IV population moment uniquely identifies β_{IV}.

Finally, if $\delta_{\text{IV}} = 0_p$, then $Q_{\text{IV}}(\beta_{\text{IV}} - \beta) = 0_p$. Nonsingularity of Q_{IV} implies $\beta_{\text{IV}} = \beta$. Hence,

when the projected IV moment bias vanishes, the projected IV population moment identifies the structural coefficient β.

Define $r_{\text{IV},i} := y_i - x_i^\top \beta_{\text{IV}} = \varsigma_i - x_i^\top Q_{\text{IV}}^{-1}\delta_{\text{IV}}$, and write $r_{\text{IV}} := (r_{\text{IV},1}, \ldots, r_{\text{IV},n})^\top = Y - X\beta_{\text{IV}}$. Then

$$\mathbb{E}[(\Pi_z x_i) r_{\text{IV},i}] = \delta_{\text{IV}} - Q_{\text{IV}} Q_{\text{IV}}^{-1}\delta_{\text{IV}} = 0_p.$$

Hence, β_{IV} satisfies the population projected normal equation

$$\Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i r_{\text{IV},i}) = 0_p. \tag{C.2}$$

Importantly, (C.2) does not generally imply $\mathbb{E}(z_i r_{\text{IV},i}) = 0_q$.

Using $Y = X\beta_{\text{IV}} + r_{\text{IV}}$, the 2SLS estimator satisfies the exact decomposition

$$\widehat{\beta}_{\text{IV}} - \beta_{\text{IV}} = \left(\frac{1}{n}X^\top P_Z X\right)^{-1}\left(\frac{1}{n}X^\top P_Z r_{\text{IV}}\right). \tag{C.3}$$

(ii) Consistency. Suppose Assumption 1 (i) holds. We first consider the denominator in (C.3).

Under the maintained second-moment conditions, the Cauchy–Schwarz inequality gives $\mathbb{E}\|x_i z_i^\top\|_{\text{F}} \le \{\mathbb{E}\|x_i\|_2^2\}^{1/2}\{\mathbb{E}\|z_i\|_2^2\}^{1/2} < \infty$, while $\mathbb{E}\|z_i z_i^\top\|_{\text{F}} = \mathbb{E}\|z_i\|_2^2 < \infty$. Therefore, the weak law of large numbers applied entrywise gives $\frac{1}{n}X^\top Z \xrightarrow{\text{p}} \Sigma_{xz}$, $\frac{1}{n}Z^\top Z \xrightarrow{\text{p}} \Sigma_z$, and $\frac{1}{n}Z^\top X \xrightarrow{\text{p}} \Sigma_{zx}$. Since $\Sigma_z \in \mathbb{S}_{++}^q$, the continuous mapping theorem gives $(\frac{1}{n}Z^\top Z)^{-1} \xrightarrow{\text{p}} \Sigma_z^{-1}$. Thus,

$$\frac{1}{n}X^\top P_Z X = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\left(\frac{1}{n}Z^\top X\right) \xrightarrow{\text{p}} Q_{\text{IV}}. \tag{C.4}$$

Because $Q_{\text{IV}} \in \mathbb{S}_{++}^p$, $(\frac{1}{n}X^\top P_Z X)^{-1} \xrightarrow{\text{p}} Q_{\text{IV}}^{-1}$.

We next consider the numerator. Since $r_{\text{IV},i} = \varsigma_i - x_i^\top Q_{\text{IV}}^{-1}\delta_{\text{IV}}$, we have $\mathbb{E}(r_{\text{IV},i}^2) \le 2\mathbb{E}(\varsigma_i^2) + 2\|Q_{\text{IV}}^{-1}\delta_{\text{IV}}\|_2^2\mathbb{E}\|x_i\|_2^2 < \infty$. It follows from Cauchy–Schwarz that $\mathbb{E}\|z_i r_{\text{IV},i}\|_2 < \infty$, and hence $\frac{1}{n}\sum_{i=1}^n z_i r_{\text{IV},i} \xrightarrow{\text{p}} \mathbb{E}(z_i r_{\text{IV},i})$. Therefore,

$$\frac{1}{n}X^\top P_Z r_{\text{IV}} = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\frac{1}{n}\sum_{i=1}^n z_i r_{\text{IV},i} \xrightarrow{\text{p}} \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i r_{\text{IV},i}) = 0_p,$$

where the last equality follows from (C.2).

Combining this result with (C.4) and (C.3) gives $\widehat{\beta}_{\text{IV}} - \beta_{\text{IV}} \xrightarrow{\text{p}} 0_p$. Hence, $\widehat{\beta}_{\text{IV}} \xrightarrow{\text{p}} \beta_{\text{IV}}$.

(iii) Asymptotic normality. Suppose further that Assumption 1 (ii) holds. Multiplying (C.3) by $\sqrt{n}$ gives

$$\sqrt{n}(\widehat{\beta}_{\text{IV}} - \beta_{\text{IV}}) = \left(\frac{1}{n}X^\top P_Z X\right)^{-1}\left(\frac{1}{\sqrt{n}}X^\top P_Z r_{\text{IV}}\right). \tag{C.5}$$

By (C.4), the first factor converges in probability to Q_{IV}^{-1}. It therefore remains to derive the limiting distribution of the normalized numerator.

Directly expanding the numerator gives

$$\frac{1}{\sqrt{n}}X^\top P_Z r_{\text{IV}} = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n z_i r_{\text{IV},i}. \tag{C.6}$$

The projected normal equation (C.2) only requires $\Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i r_{\text{IV},i}) = 0_p$, while $\mathbb{E}(z_i r_{\text{IV},i})$ itself need not vanish. Consequently, $n^{-1/2}\sum_{i=1}^n z_i r_{\text{IV},i}$ may contain a mean component of order $\sqrt{n}$, so this sum is not generally $O_p(1)$. Thus, unlike a directly centered CLT term, the random projection coefficient in (C.6) cannot immediately be replaced by its probability limit.

To obtain a centered term, define

$$e_{\text{IV},i} := r_{\text{IV},i} - z_i^\top \Sigma_z^{-1} \mathbb{E}(z_i r_{\text{IV},i}).$$

The consistency argument gives $\mathbb{E}\|z_i r_{\text{IV},i}\|_2 < \infty$, so $\mathbb{E}(z_i r_{\text{IV},i})$ is finite. By construction,

$$\mathbb{E}(z_i e_{\text{IV},i}) = \mathbb{E}(z_i r_{\text{IV},i}) - \Sigma_z \Sigma_z^{-1} \mathbb{E}(z_i r_{\text{IV},i}) = 0_q.$$

Assumption 1 (ii) further gives $\mathbb{E}\|z_i e_{\text{IV},i}\|_2^2 < \infty$. Hence, the multivariate central limit theorem implies $n^{-1/2} \sum_{i=1}^n z_i e_{\text{IV},i} = O_p(1)$.

Writing $e_{\text{IV}} := (e_{\text{IV},1}, \ldots, e_{\text{IV},n})^\top$, the preceding definition gives $r_{\text{IV}} = e_{\text{IV}} + Z\Sigma_z^{-1}\mathbb{E}(z_i r_{\text{IV},i})$. At this point, the IV argument differs from the corresponding SV argument. For SV, the residual-maker eliminates the ancillary-variable component because $P_W^\perp W = 0$. For IV, however, $P_Z Z = Z$, so the component in the span of Z is retained. Substituting the decomposition of r_{IV} into (C.6) therefore gives the exact decomposition

$$\begin{aligned} \frac{1}{\sqrt{n}} X^\top P_Z r_{\text{IV}} = \left(\frac{1}{n} X^\top Z\right) \left(\frac{1}{n} Z^\top Z\right)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n z_i e_{\text{IV},i} \\ + \frac{1}{\sqrt{n}} \sum_{i=1}^n x_i z_i^\top \Sigma_z^{-1} \mathbb{E}(z_i r_{\text{IV},i}). \end{aligned} \tag{C.7}$$

Thus, in contrast to SV, recentering does not remove the ancillary-variable component but instead produces an additional first-order correction term.

For the first term in (C.7), the consistency argument gives

$$\left(\frac{1}{n} X^\top Z\right) \left(\frac{1}{n} Z^\top Z\right)^{-1} = \Sigma_{xz} \Sigma_z^{-1} + o_p(1).$$

Since $n^{-1/2}\sum_{i=1}^n z_i e_{\mathrm{IV},i} = O_p(1)$, the estimation error in this coefficient contributes only $o_p(1)$. Hence,

$$\left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n z_i e_{\mathrm{IV},i} = \frac{1}{\sqrt{n}}\sum_{i=1}^n(\Pi_z x_i)e_{\mathrm{IV},i} + o_p(1).$$

Substituting this expression into (C.7) motivates defining

$$\psi_{\mathrm{IV},i} := (\Pi_z x_i)e_{\mathrm{IV},i} + x_i z_i^\top \Sigma_z^{-1}\mathbb{E}(z_i r_{\mathrm{IV},i}). \tag{C.8}$$

The normalized numerator then admits the first-order representation

$$\frac{1}{\sqrt{n}}X^\top P_Z r_{\mathrm{IV}} = \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{IV},i} + o_p(1). \tag{C.9}$$

It remains to verify that the summands in (C.9) are centered and have finite second moments. Using $\mathbb{E}(z_i e_{\mathrm{IV},i}) = 0_q$,

$$\mathbb{E}(\psi_{\mathrm{IV},i}) = \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i e_{\mathrm{IV},i}) + \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i r_{\mathrm{IV},i}) = 0_p,$$

where the last equality follows from (C.2).

For the second moment, $\mathbb{E}\|(\Pi_z x_i)e_{\mathrm{IV},i}\|_2^2 \le \|\Sigma_{xz}\Sigma_z^{-1}\|_{\mathrm{op}}^2\mathbb{E}\|z_i e_{\mathrm{IV},i}\|_2^2 < \infty$. Moreover,

$$\mathbb{E}\left\|x_i z_i^\top \Sigma_z^{-1}\mathbb{E}(z_i r_{\mathrm{IV},i})\right\|_2^2 \le \left\|\Sigma_z^{-1}\mathbb{E}(z_i r_{\mathrm{IV},i})\right\|_2^2\mathbb{E}\|x_i z_i^\top\|_{\mathrm{op}}^2 < \infty.$$

Therefore, $\mathbb{E}\|\psi_{\mathrm{IV},i}\|_2^2 < \infty$. The second bound is precisely where the stronger condition $\mathbb{E}\|x_i z_i^\top\|_{\mathrm{op}}^2 < \infty$ in Assumption 1(ii) is needed. Under valid IV, $\mathbb{E}(z_i r_{\mathrm{IV},i}) = 0_q$, so the correction term in (C.8) vanishes and this additional bound is unnecessary for that term.

Because the observations are i.i.d., the $\psi_{\text{IV},i}$ are i.i.d., centered, and have finite second moments. Therefore, the multivariate central limit theorem gives

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_{\text{IV},i} \xrightarrow{\text{d}} \mathcal{N}(0,\Sigma_{\text{IV}}), \qquad \Sigma_{\text{IV}} := \text{Var}(\psi_{\text{IV},i}). \tag{C.10}$$

Combining (C.9) and (C.10) yields $\frac{1}{\sqrt{n}}X^\top P_Z r_{\text{IV}} \xrightarrow{\text{d}} \mathcal{N}(0,\Sigma_{\text{IV}})$.

Finally, combining this result with (C.5), (C.4), and Slutsky's theorem gives

$$\sqrt{n}(\widehat{\beta}_{\text{IV}} - \beta_{\text{IV}}) \xrightarrow{\text{d}} \mathcal{N}\left(0, Q_{\text{IV}}^{-1}\Sigma_{\text{IV}}Q_{\text{IV}}^{-1}\right) = \mathcal{N}(0, V_{\text{IV}}), \tag{C.11}$$

where $V_{\text{IV}} := Q_{\text{IV}}^{-1}\Sigma_{\text{IV}}Q_{\text{IV}}^{-1}$. This completes the proof.

□

C.2 Proof of Theorem 2

Proof of Theorem 2. Recall that $\Pi_w^\perp x_i = x_i - \Sigma_{xw}\Sigma_w^{-1}w_i$, $Q_{\text{SV}} = \mathbb{E}\{(\Pi_w^\perp x_i)x_i^\top\}$, $\delta_{\text{SV}} = \mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\}$, and $\beta_{\text{SV}} = \beta + Q_{\text{SV}}^{-1}\delta_{\text{SV}}$.

(i) Identification. Recall that the residualized population moment condition is $\mathbb{E}[(\Pi_w^\perp x_i)(y_i - x_i^\top\theta)] = 0_p$, where $\theta \in \mathbb{R}^p$ is a candidate coefficient vector. Using $y_i = x_i^\top\beta + \varsigma_i$, we have

$$\mathbb{E}\Big[(\Pi_w^\perp x_i)\big(y_i - x_i^\top\theta\big)\Big] = \mathbb{E}\Big[(\Pi_w^\perp x_i)x_i^\top\Big](\beta - \theta) + \mathbb{E}\Big[(\Pi_w^\perp x_i)\varsigma_i\Big].$$

By the definitions $Q_{\text{SV}} := \mathbb{E}[(\Pi_w^\perp x_i)x_i^\top]$ and $\delta_{\text{SV}} := \mathbb{E}[(\Pi_w^\perp x_i)\varsigma_i]$, the population moment condition is therefore equivalent to

$$Q_{\text{SV}}(\theta - \beta) = \delta_{\text{SV}}. \tag{C.12}$$

By Assumption 2(i), $Q_{\mathrm{SV}} \in \mathbb{S}^p_{++}$, and hence Q_{SV} is nonsingular. Left-multiplying (C.12) by Q_{SV}^{-1} gives

$$\theta = \beta + Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}} = \beta_{\mathrm{SV}}.$$

Thus a solution exists.

To see that the solution is unique, suppose that both θ_1 and θ_2 satisfy (C.12). Then $Q_{\mathrm{SV}}(\theta_1 - \theta_2) = 0_p$. Since Q_{SV} is nonsingular, its null space contains only 0_p, so $\theta_1 - \theta_2 = 0_p$, and hence $\theta_1 = \theta_2$. Therefore, the residualized population moment uniquely identifies β_{SV}.

Finally, if $\delta_{\mathrm{SV}} = 0_p$, then $Q_{\mathrm{SV}}(\beta_{\mathrm{SV}} - \beta) = 0_p$. Nonsingularity of Q_{SV} implies $\beta_{\mathrm{SV}} = \beta$. Hence, when the residualized moment bias vanishes, the surrogate-control population moment identifies the structural coefficient β.

Within the remainder of the proof, define $\Gamma_w := \Sigma_w^{-1}\Sigma_{wx}$, $\widetilde{x}_i := x_i - \Gamma_w^\top w_i = \Pi_w^\perp x_i$, and let $\widetilde{X}$ denote the matrix whose i-th row is $\widetilde{x}_i^\top$. Then

$$x_i = \Gamma_w^\top w_i + \widetilde{x}_i, \qquad X = W\Gamma_w + \widetilde{X}. \tag{C.13}$$

By the definition of the population linear projection,

$$\mathbb{E}(\widetilde{x}_i w_i^\top) = \Sigma_{xw} - \Gamma_w^\top \Sigma_w = 0_{p\times m}. \tag{C.14}$$

Consequently,

$$Q_{\mathrm{SV}} = \mathbb{E}(\widetilde{x}_i x_i^\top) = \mathbb{E}(\widetilde{x}_i \widetilde{x}_i^\top) + \mathbb{E}(\widetilde{x}_i w_i^\top)\Gamma_w = \mathbb{E}(\widetilde{x}_i \widetilde{x}_i^\top).$$

Hence Q_{SV} is symmetric, and it is positive definite by Assumption 2(i).

Define $r_{\mathrm{SV},i} := y_i - x_i^\top \beta_{\mathrm{SV}} = \varsigma_i - x_i^\top Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}$, and write $r_{\mathrm{SV}} := (r_{\mathrm{SV},1}, \ldots, r_{\mathrm{SV},n})^\top$. Using $\widetilde{x}_i = \Pi_w^\perp x_i$, we obtain

$$\begin{aligned}\mathbb{E}(\widetilde{x}_i r_{\mathrm{SV},i}) &= \mathbb{E}(\widetilde{x}_i \varsigma_i) - \mathbb{E}(\widetilde{x}_i x_i^\top) Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}} \\ &= \delta_{\mathrm{SV}} - Q_{\mathrm{SV}} Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}} \\ &= 0_p.\end{aligned} \tag{C.15}$$

Thus, β_{SV} satisfies the population residualized normal equation.

Since $P_W^\perp W = 0$, decomposition (C.13) gives $P_W^\perp X = P_W^\perp \widetilde{X}$. Because $P_W^\perp$ is symmetric, $X^\top P_W^\perp X = \widetilde{X}^\top P_W^\perp \widetilde{X}$ and $X^\top P_W^\perp r_{\mathrm{SV}} = \widetilde{X}^\top P_W^\perp r_{\mathrm{SV}}$. The estimator therefore satisfies the exact decomposition

$$\widehat{\beta}_{\mathrm{SV}} - \beta_{\mathrm{SV}} = \left(\frac{1}{n}\widetilde{X}^\top P_W^\perp \widetilde{X}\right)^{-1} \left(\frac{1}{n}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}}\right). \tag{C.16}$$

(ii) Consistency. Suppose Assumption 2 (i) holds. The maintained second-moment conditions and the weak law of large numbers give $\frac{1}{n}\widetilde{X}^\top \widetilde{X} \xrightarrow{\mathrm{p}} \mathbb{E}(\widetilde{x}_i \widetilde{x}_i^\top) = Q_{\mathrm{SV}}$, $\frac{1}{n}\widetilde{X}^\top W \xrightarrow{\mathrm{p}} \mathbb{E}(\widetilde{x}_i w_i^\top) = 0_{p\times m}$, and $\frac{1}{n}W^\top W \xrightarrow{\mathrm{p}} \Sigma_w$. Since $\Sigma_w \in \mathbb{S}_{++}^m$,

$$\begin{aligned}\frac{1}{n}\widetilde{X}^\top P_W^\perp \widetilde{X} &= \frac{1}{n}\widetilde{X}^\top \widetilde{X} - \left(\frac{1}{n}\widetilde{X}^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\left(\frac{1}{n}W^\top \widetilde{X}\right) \\ &\xrightarrow{\mathrm{p}} Q_{\mathrm{SV}}.\end{aligned}$$

It follows that $\{\frac{1}{n}\widetilde{X}^\top P_W^\perp \widetilde{X}\}^{-1} \xrightarrow{\mathrm{p}} Q_{\mathrm{SV}}^{-1}$.

Similarly, the weak law of large numbers gives $\frac{1}{n}\sum_{i=1}^n \widetilde{x}_i r_{\mathrm{SV},i} \xrightarrow{\mathrm{p}} \mathbb{E}(\widetilde{x}_i r_{\mathrm{SV},i}) = 0_p$ by (C.15), and $\frac{1}{n}\sum_{i=1}^n w_i r_{\mathrm{SV},i} \xrightarrow{\mathrm{p}} \mathbb{E}(w_i r_{\mathrm{SV},i})$. Therefore,

$$\begin{aligned}\frac{1}{n}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}} &= \frac{1}{n}\sum_{i=1}^n \widetilde{x}_i r_{\mathrm{SV},i} - \left(\frac{1}{n}\widetilde{X}^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^n w_i r_{\mathrm{SV},i}\right) \\ &\xrightarrow{\mathrm{p}} 0_p.\end{aligned}$$

Combining these limits with (C.16) yields

$$\widehat{\beta}_{\mathrm{SV}} \xrightarrow{\mathrm{p}} \beta_{\mathrm{SV}}.$$

(iii) Asymptotic normality. Suppose further that Assumption 2 (ii) holds. Multiplying (C.16) by $\sqrt{n}$ gives

$$\sqrt{n}(\widehat{\beta}_{\mathrm{SV}} - \beta_{\mathrm{SV}}) = \left(\frac{1}{n}\widetilde{X}^\top P_W^\perp \widetilde{X}\right)^{-1} \frac{1}{\sqrt{n}}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}}.$$

The first factor converges in probability to Q_{SV}^{-1}, so it remains to derive the limiting distribution of the normalized numerator.

A direct expansion gives

$$\frac{1}{\sqrt{n}}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}} = \frac{1}{\sqrt{n}}\sum_{i=1}^n \widetilde{x}_i r_{\mathrm{SV},i} - \left(\frac{1}{n}\widetilde{X}^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n w_i r_{\mathrm{SV},i}. \tag{C.17}$$

Although $\mathbb{E}(\widetilde{x}_i r_{\mathrm{SV},i}) = 0_p$, this does not imply $\mathbb{E}(w_i r_{\mathrm{SV},i}) = 0_m$. Hence the last sum in (C.17) may contain a nonzero mean component of order $\sqrt{n}$, so the second term cannot yet be treated as negligible.

To center this sum, define

$$c_{\mathrm{SV}} := \Sigma_w^{-1}\mathbb{E}(w_i r_{\mathrm{SV},i}), \qquad e_{\mathrm{SV},i} := r_{\mathrm{SV},i} - w_i^\top c_{\mathrm{SV}}. \tag{C.18}$$

Then $\mathbb{E}(w_i e_{\mathrm{SV},i}) = \mathbb{E}(w_i r_{\mathrm{SV},i}) - \Sigma_w c_{\mathrm{SV}} = 0_m$. Writing $e_{\mathrm{SV}} := (e_{\mathrm{SV},1}, \ldots, e_{\mathrm{SV},n})^\top$, equation (C.18) gives $r_{\mathrm{SV}} = e_{\mathrm{SV}} + W c_{\mathrm{SV}}$. Since $P_W^\perp W = 0$, we have the exact identity

$$P_W^\perp r_{\mathrm{SV}} = P_W^\perp e_{\mathrm{SV}}.$$

The corrected numerator therefore satisfies

$$\frac{1}{\sqrt{n}}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}} = \frac{1}{\sqrt{n}}\sum_{i=1}^n \widetilde{x}_i e_{\mathrm{SV},i} - \left(\frac{1}{n}\widetilde{X}^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n w_i e_{\mathrm{SV},i}.$$

Assumption 2(ii) and $\mathbb{E}(w_i e_{\mathrm{SV},i}) = 0_m$ imply $n^{-1/2}\sum_{i=1}^n w_i e_{\mathrm{SV},i} = O_p(1)$ by the multivariate central limit theorem. Moreover, $\frac{1}{n}\widetilde{X}^\top W = o_p(1)$ and $\{\frac{1}{n}W^\top W\}^{-1} = O_p(1)$. Thus, the second term is $o_p(1)$.

Define $\psi_{\mathrm{SV},i} := \widetilde{x}_i e_{\mathrm{SV},i} = (\Pi_w^\perp x_i)e_{\mathrm{SV},i}$. The preceding argument gives the first-order representation

$$\frac{1}{\sqrt{n}}X^\top P_W^\perp r_{\mathrm{SV}} = \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{SV},i} + o_p(1). \tag{C.19}$$

The score is centered because, by (C.14) and (C.15),

$$\mathbb{E}(\psi_{\mathrm{SV},i}) = \mathbb{E}(\widetilde{x}_i r_{\mathrm{SV},i}) - \mathbb{E}(\widetilde{x}_i w_i^\top)c_{\mathrm{SV}} = 0_p.$$

It remains to verify that $\mathbb{E}\|\psi_{\mathrm{SV},i}\|_2^2 < \infty$. Since $r_{\mathrm{SV},i} = \varsigma_i - x_i^\top Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}$,

$$\mathbb{E}\|x_i r_{\mathrm{SV},i}\|_2^2 \le 2\mathbb{E}\|x_i\varsigma_i\|_2^2 + 2\|Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}\|_2^2\mathbb{E}\|x_i x_i^\top\|_{\mathrm{op}}^2 < \infty.$$

The consistency argument and the nonsingularity of Σ_w imply that c_{SV} in (C.18) is finite. Hence,

$$\mathbb{E}\|x_i e_{\mathrm{SV},i}\|_2^2 \le 2\mathbb{E}\|x_i r_{\mathrm{SV},i}\|_2^2 + 2\|c_{\mathrm{SV}}\|_2^2\mathbb{E}\|x_i w_i^\top\|_{\mathrm{op}}^2 < \infty.$$

Finally, $\psi_{\mathrm{SV},i} = x_i e_{\mathrm{SV},i} - \Gamma_w^\top w_i e_{\mathrm{SV},i}$, so

$$\mathbb{E}\|\psi_{\mathrm{SV},i}\|_2^2 \le 2\mathbb{E}\|x_i e_{\mathrm{SV},i}\|_2^2 + 2\|\Gamma_w^\top\|_{\mathrm{op}}^2\mathbb{E}\|w_i e_{\mathrm{SV},i}\|_2^2 < \infty.$$

Because the observations are i.i.d., the random vectors $\psi_{\text{SV},i}$ are i.i.d., centered, and have finite second moments. The multivariate central limit theorem therefore gives

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_{\text{SV},i} \xrightarrow{\text{d}} \mathcal{N}(0,\Sigma_{\text{SV}}), \qquad \Sigma_{\text{SV}} := \text{Var}(\psi_{\text{SV},i}).$$

Combining this result with (C.19) yields

$$\frac{1}{\sqrt{n}}X^{\top}P_W^{\perp}r_{\text{SV}} \xrightarrow{\text{d}} \mathcal{N}(0,\Sigma_{\text{SV}}).$$

Finally, Slutsky's theorem gives

$$\sqrt{n}(\widehat{\beta}_{\text{SV}}-\beta_{\text{SV}}) \xrightarrow{\text{d}} \mathcal{N}\left(0, Q_{\text{SV}}^{-1}\Sigma_{\text{SV}}Q_{\text{SV}}^{-1}\right).$$

Thus,

$$V_{\text{SV}} = Q_{\text{SV}}^{-1}\Sigma_{\text{SV}}Q_{\text{SV}}^{-1}.$$

This completes the proof.

□

C.3 Proof of Proposition 3

Proof of Proposition 3. Recall that $h_i = \{(\Pi_z x_i)^{\top}, (\Pi_w^{\perp}x_i)^{\top}\}^{\top}$ and $J = \mathbb{E}(h_i x_i^{\top}) = \{Q_{\text{IV}}^{\top}, Q_{\text{SV}}^{\top}\}^{\top}$. Under exact projected validity, $\mathbb{E}(h_i\varsigma_i) = 0_{2p}$.

Population moment equation. Let $\theta \in \mathbb{R}^p$ be a candidate coefficient vector. Using $y_i = x_i^{\top}\beta + \varsigma_i$, the stacked population moment satisfies

$$\mathbb{E}\{h_i(y_i - x_i^{\top}\theta)\} = J(\beta-\theta) + \mathbb{E}(h_i\varsigma_i) = J(\beta-\theta).$$

Hence, θ satisfies the stacked moment restriction if and only if $J(\theta - \beta) = 0_{2p}$. Therefore, β is the unique solution if and only if $\ker(J) = \{0_p\}$, which is equivalent to $\operatorname{rank}(J) = p$.

Rank characterization. Because J stacks the two projected Jacobian blocks, $Jv = 0_{2p}$ for $v \in \mathbb{R}^p$ if and only if $Q_{\text{IV}}v = 0_p$ and $Q_{\text{SV}}v = 0_p$. It follows that

$$\ker(J) = \ker(Q_{\text{IV}}) \cap \ker(Q_{\text{SV}}).$$

Consequently, $\operatorname{rank}(J) = p$ if and only if $\ker(Q_{\text{IV}}) \cap \ker(Q_{\text{SV}}) = \{0_p\}$, proving (3.2).

Complementarity. The preceding characterization does not require either $\ker(Q_{\text{IV}})$ or $\ker(Q_{\text{SV}})$ to be trivial individually. Thus, either projected moment block may be rank deficient as long as no nonzero parameter direction belongs to both null spaces. Equivalently, directions that are not identified by one block may still be identified by the other, and the stacked system uniquely identifies β whenever their unidentified directions have trivial intersection.

□

C.4 Proof of Theorem 4

Proof of Theorem 4. Recall that $\varsigma_i = y_i - x_i^\top \beta$ and

$$\widehat{g}(\theta) = \frac{1}{n} \begin{pmatrix} X^\top P_Z (Y - X\theta) \\ \\ X^\top P_W^\perp (Y - X\theta) \end{pmatrix}.$$

Let

$$\widehat{J}_n := \frac{1}{n} \begin{pmatrix} X^\top P_Z X \\ \\ X^\top P_W^\perp X \end{pmatrix}.$$

Since the sample moment is linear in θ,

$$\widehat{g}(\theta) = \widehat{g}(\beta) - \widehat{J}_n(\theta - \beta). \tag{C.20}$$

Assumption 3 (i) gives $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$ and $\mathbb{E}\{(\Pi_w^{\perp} x_i)\varsigma_i\} = 0_p$. Thus, both projected moment blocks are correctly specified at β.

Identification. Let $g(\theta)$ denote the population counterpart of $\widehat{g}(\theta)$. By linearity and the projected moment restrictions,

$$g(\theta) = -J(\theta - \beta).$$

Hence $g(\theta) = 0_{2p}$ if and only if $J(\theta - \beta) = 0_{2p}$. Since Assumption 3(i) gives $\operatorname{rank}(J) = p$, the null space of J is trivial. Therefore, $\theta = \beta$, so β is the unique solution to the population combined projected moment condition.

The first-order condition for $\widehat{\beta}(M)$ is $\widehat{J}_n^{\top} M \widehat{g}\{\widehat{\beta}(M)\} = 0_p$. Using (C.20), we obtain the exact estimation-error decomposition

$$\widehat{\beta}(M) - \beta = \left(\widehat{J}_n^{\top} M \widehat{J}_n\right)^{-1} \widehat{J}_n^{\top} M \widehat{g}(\beta). \tag{C.21}$$

Consistency. Suppose Assumption 3 (i) holds. We first establish the probability limit of $\widehat{J}_n$.

For the IV block, the maintained second-moment conditions and the weak law of large numbers give $\frac{1}{n} X^{\top} Z \xrightarrow{\mathrm{p}} \Sigma_{xz}$, $\frac{1}{n} Z^{\top} Z \xrightarrow{\mathrm{p}} \Sigma_z$, and $\frac{1}{n} Z^{\top} X \xrightarrow{\mathrm{p}} \Sigma_{zx}$. Since $\Sigma_z \in \mathbb{S}_{++}^q$, $(\frac{1}{n} Z^{\top} Z)^{-1} \xrightarrow{\mathrm{p}}$

Σ_z^{-1}. Hence,

$$\frac{1}{n}X^\top P_Z X = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\left(\frac{1}{n}Z^\top X\right) \xrightarrow{\text{p}} Q_{\text{IV}}.$$

For the SV block, the weak law of large numbers gives $\frac{1}{n}X^\top X \xrightarrow{\text{p}} \Sigma_x$, $\frac{1}{n}X^\top W \xrightarrow{\text{p}} \Sigma_{xw}$, $\frac{1}{n}W^\top W \xrightarrow{\text{p}} \Sigma_w$, and $\frac{1}{n}W^\top X \xrightarrow{\text{p}} \Sigma_{wx}$. Since $\Sigma_w \in \mathbb{S}^m_{++}$, $(\frac{1}{n}W^\top W)^{-1} \xrightarrow{\text{p}} \Sigma_w^{-1}$. Therefore,

$$\frac{1}{n}X^\top P_W^\perp X \xrightarrow{\text{p}} \Sigma_x - \Sigma_{xw}\Sigma_w^{-1}\Sigma_{wx} = Q_{\text{SV}}.$$

Combining the two blocks gives

$$\widehat{J}_n \xrightarrow{\text{p}} \begin{pmatrix} Q_{\text{IV}} \\ \\ Q_{\text{SV}} \end{pmatrix} = J. \tag{C.22}$$

We next establish the probability limit of $\widehat{g}(\beta)$. For the IV block,

$$\frac{1}{n}X^\top P_Z \varsigma = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^n z_i\varsigma_i\right) \xrightarrow{\text{p}} \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i) = 0_p,$$

where the final equality follows from $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$. In particular, this does not require the stronger raw IV restriction $\mathbb{E}(z_i\varsigma_i) = 0_q$.

For the SV block,

$$\frac{1}{n}X^\top P_W^\perp \varsigma \xrightarrow{\text{p}} \mathbb{E}(x_i\varsigma_i) - \Sigma_{xw}\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i) = \mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} = 0_p.$$

Hence,

$$\widehat{g}(\beta) \xrightarrow{\text{p}} 0_{2p}. \tag{C.23}$$

Since $M \in \mathbb{S}^{2p}_{++}$ and J has full column rank p, $J^\top M J \in \mathbb{S}^p_{++}$. Indeed, for every nonzero $c \in \mathbb{R}^p$, $c^\top J^\top M J c = (Jc)^\top M (Jc) > 0$. It follows from (C.22) and the continuous mapping

theorem that

$$\left(\widehat{J}_n^\top M \widehat{J}_n\right)^{-1} \widehat{J}_n^\top M \xrightarrow{\mathrm{p}} (J^\top M J)^{-1} J^\top M = K_M.$$

Combining this result with (C.21) and (C.23) gives $\widehat{\beta}(M) - \beta \xrightarrow{\mathrm{p}} 0_p$. Therefore, $\widehat{\beta}(M) \xrightarrow{\mathrm{p}} \beta$.

Asymptotic normality. Suppose further that Assumption 3 (ii) holds. Multiplying (C.21) by $\sqrt{n}$ gives

$$\sqrt{n}\{\widehat{\beta}(M) - \beta\} = \left(\widehat{J}_n^\top M \widehat{J}_n\right)^{-1} \widehat{J}_n^\top M \sqrt{n}\, \widehat{g}(\beta). \tag{C.24}$$

By the consistency argument, the matrix multiplying $\sqrt{n}\,\widehat{g}(\beta)$ converges to K_M. It therefore remains to derive a central limit theorem for the two projected moment blocks in $\sqrt{n}\,\widehat{g}(\beta)$.

For notational convenience within this proof, define $c_z := \Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)$ and $c_w := \Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)$, and let $e_{z,i} := \varsigma_i - z_i^\top c_z$ and $e_{w,i} := \varsigma_i - w_i^\top c_w$. The standing second-moment conditions imply that $\mathbb{E}(z_i\varsigma_i)$ and $\mathbb{E}(w_i\varsigma_i)$ are finite, so c_z and c_w are finite nonrandom vectors. By construction, $\mathbb{E}(z_i e_{z,i}) = 0_q$ and $\mathbb{E}(w_i e_{w,i}) = 0_m$.

We first consider the projected IV block. A direct expansion in terms of ς_i cannot generally be treated by replacing the sample projection coefficient by its probability limit, because the projected restriction $\Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i) = 0_p$ does not imply $\mathbb{E}(z_i\varsigma_i) = 0_q$. Thus, $n^{-1/2}\sum_{i=1}^n z_i\varsigma_i$ may contain a mean component of order $\sqrt{n}$.

Using $\varsigma_i = e_{z,i} + z_i^\top c_z$, the IV block admits the exact decomposition

$$\begin{aligned}\frac{1}{\sqrt{n}} X^\top P_Z \varsigma = \left(\frac{1}{n} X^\top Z\right)\left(\frac{1}{n} Z^\top Z\right)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n z_i e_{z,i} \\ + \frac{1}{\sqrt{n}} \sum_{i=1}^n x_i z_i^\top c_z.\end{aligned} \tag{C.25}$$

The second term remains because $P_Z Z = Z$; hence, the component of the structural error lying in the span of Z is retained by the IV projection.

Assumption 3 (ii) gives $\mathbb{E}\|z_i e_{z,i}\|_2^2 < \infty$, so $n^{-1/2}\sum_{i=1}^n z_i e_{z,i} = O_p(1)$. Moreover,

$$\left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1} = \Sigma_{xz}\Sigma_z^{-1} + o_p(1).$$

Therefore, the first term in (C.25) equals $n^{-1/2}\sum_{i=1}^n(\Pi_z x_i)e_{z,i} + o_p(1)$, and hence

$$\frac{1}{\sqrt{n}}X^\top P_Z \varsigma = \frac{1}{\sqrt{n}}\sum_{i=1}^n\left\{(\Pi_z x_i)e_{z,i} + x_i z_i^\top c_z\right\} + o_p(1). \tag{C.26}$$

We next consider the projected SV block. Here the same recentering has a different consequence. Since $\varsigma_i = e_{w,i} + w_i^\top c_w$ and $P_W^\perp W = 0$, the component lying in the span of W is removed exactly:

$$\frac{1}{\sqrt{n}}X^\top P_W^\perp \varsigma = \frac{1}{\sqrt{n}}X^\top P_W^\perp e_w,$$

where $e_w = (e_{w,1}, \ldots, e_{w,n})^\top$.

Let $\Pi_w^\perp X$ denote the matrix whose i-th row is $(\Pi_w^\perp x_i)^\top$. Since $X = \Pi_w^\perp X + W\Sigma_w^{-1}\Sigma_{wx}$, the identity $P_W^\perp W = 0$ implies $P_W^\perp X = P_W^\perp(\Pi_w^\perp X)$. Expanding the sample residual-maker therefore gives

$$\begin{aligned}\frac{1}{\sqrt{n}}X^\top P_W^\perp \varsigma = {} & \frac{1}{\sqrt{n}}\sum_{i=1}^n(\Pi_w^\perp x_i)e_{w,i} \\ & - \left\{\frac{1}{n}\sum_{i=1}^n(\Pi_w^\perp x_i)w_i^\top\right\}\left(\frac{1}{n}\sum_{i=1}^n w_i w_i^\top\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n w_i e_{w,i}.\end{aligned} \tag{C.27}$$

By the definition of the population residualization, $\mathbb{E}\{(\Pi_w^\perp x_i)w_i^\top\} = 0_{p\times m}$. The weak law of large numbers therefore gives $n^{-1}\sum_{i=1}^n(\Pi_w^\perp x_i)w_i^\top = o_p(1)$. Also, $(n^{-1}\sum_{i=1}^n w_i w_i^\top)^{-1} \xrightarrow{\mathrm{P}} \Sigma_w^{-1}$,

while $\mathbb{E}(w_i e_{w,i}) = 0_m$ and $\mathbb{E}\|w_i e_{w,i}\|_2^2 < \infty$ imply $n^{-1/2}\sum_{i=1}^n w_i e_{w,i} = O_p(1)$. Hence, the second term in (C.27) is $o_p(1)$, so

$$\frac{1}{\sqrt{n}} X^\top P_W^\perp \varsigma = \frac{1}{\sqrt{n}} \sum_{i=1}^n (\Pi_w^\perp x_i) e_{w,i} + o_p(1). \tag{C.28}$$

Equations (C.26) and (C.28) collect the first-order variation of the two feasible projected moment blocks. Recall from Section 3.1 that $\psi_{\mathrm{IV},i}$ and $\psi_{\mathrm{SV},i}$ denote the corresponding first-order scores. Under the notation introduced above, these previously defined scores admit the equivalent representations

$$\psi_{\mathrm{IV},i} = (\Pi_z x_i) e_{z,i} + x_i z_i^\top c_z, \qquad \psi_{\mathrm{SV},i} = (\Pi_w^\perp x_i) e_{w,i}.$$

Recall also that $\psi_i = (\psi_{\mathrm{IV},i}^\top, \psi_{\mathrm{SV},i}^\top)^\top$ and $\Sigma = \mathrm{Var}(\psi_i)$. Stacking (C.26) and (C.28) gives the first-order representation

$$\sqrt{n}\,\widehat{g}(\beta) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_i + o_p(1). \tag{C.29}$$

It remains to verify that the summands in (C.29) are centered and have finite second moments.

For the IV component, using $\mathbb{E}(z_i e_{z,i}) = 0_q$,

$$\mathbb{E}(\psi_{\mathrm{IV},i}) = \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i e_{z,i}) + \Sigma_{xz} c_z = \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i \varsigma_i) = 0_p,$$

where the last equality follows from the projected IV restriction. For the SV component,

$$\mathbb{E}(\psi_{\mathrm{SV},i}) = \mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} - \mathbb{E}\{(\Pi_w^\perp x_i) w_i^\top\} c_w = 0_p.$$

Hence,

$$\mathbb{E}(\psi_i) = \begin{pmatrix} \mathbb{E}(\psi_{\mathrm{IV},i}) \\ \\ \mathbb{E}(\psi_{\mathrm{SV},i}) \end{pmatrix} = \begin{pmatrix} 0_p \\ \\ 0_p \end{pmatrix} = 0_{2p}.$$

For the IV second moment, $\mathbb{E}\|(\Pi_z x_i)e_{z,i}\|_2^2 \le \|\Sigma_{xz}\Sigma_z^{-1}\|_{\mathrm{op}}^2 \mathbb{E}\|z_i e_{z,i}\|_2^2 < \infty$. Moreover, $\mathbb{E}\|x_i z_i^\top c_z\|_2^2 \le \|c_z\|_2^2 \mathbb{E}\|x_i z_i^\top\|_{\mathrm{op}}^2 < \infty$. Thus, $\mathbb{E}\|\psi_{\mathrm{IV},i}\|_2^2 < \infty$.

For the SV component, first note that $x_i e_{w,i} = x_i \varsigma_i - x_i w_i^\top c_w$. Therefore,

$$\mathbb{E}\|x_i e_{w,i}\|_2^2 \le 2\mathbb{E}\|x_i \varsigma_i\|_2^2 + 2\|c_w\|_2^2 \mathbb{E}\|x_i w_i^\top\|_{\mathrm{op}}^2 < \infty.$$

Since $(\Pi_w^\perp x_i)e_{w,i} = x_i e_{w,i} - \Sigma_{xw}\Sigma_w^{-1} w_i e_{w,i}$, we further have

$$\mathbb{E}\|\psi_{\mathrm{SV},i}\|_2^2 \le 2\mathbb{E}\|x_i e_{w,i}\|_2^2 + 2\|\Sigma_{xw}\Sigma_w^{-1}\|_{\mathrm{op}}^2 \mathbb{E}\|w_i e_{w,i}\|_2^2 < \infty.$$

Consequently, $\mathbb{E}\|\psi_i\|_2^2 < \infty$, so $\Sigma = \mathrm{Var}(\psi_i)$ is well defined.

Because the observations are i.i.d., the ψ_i are i.i.d., centered, and have finite second moments. The multivariate central limit theorem therefore gives

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_i \xrightarrow{\mathrm{d}} \mathcal{N}(0, \Sigma). \tag{C.30}$$

Combining (C.29) and (C.30) yields

$$\sqrt{n}\, \widehat{g}(\beta) \xrightarrow{\mathrm{d}} \mathcal{N}(0, \Sigma).$$

Finally, by (C.22),

$$\left(\widehat{J}_n^\top M \widehat{J}_n\right)^{-1} \widehat{J}_n^\top M \xrightarrow{\mathrm{p}} (J^\top M J)^{-1} J^\top M = K_M.$$

Combining this result with (C.24) and Slutsky's theorem gives

$$\sqrt{n}\{\widehat{\beta}(M) - \beta\} \xrightarrow{\mathrm{d}} \mathcal{N}\left(0,\, K_M \Sigma K_M^\top\right) = \mathcal{N}\{0, V(M)\}, \tag{C.31}$$

where $V(M) = K_M \Sigma K_M^\top$. By Assumption 3 (ii), $V(M) \in \mathbb{S}_{++}^p$, so the limiting distribution is nondegenerate. This completes the proof.

□

C.5 Proof of Theorem 5

Proof of Theorem 5. Recall that $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i$, $Q_{\text{IV}} = \mathbb{E}[(\Pi_z x_i) x_i^\top]$, $\Pi_w^\perp x_i = x_i - \Sigma_{xw}\Sigma_w^{-1} w_i$, and $Q_{\text{SV}} = \mathbb{E}[(\Pi_w^\perp x_i) x_i^\top]$. Define

$$\widehat{J}_n := \frac{1}{n}\begin{pmatrix} X^\top P_Z X \\ \\ X^\top P_W^\perp X \end{pmatrix}.$$

Since $\widehat{g}(\theta) = n^{-1}\big(X^\top P_Z (Y - X\theta)^\top,\, X^\top P_W^\perp (Y - X\theta)^\top\big)^\top$, the sample moment is linear in θ, so

$$\widehat{g}(\theta) = \widehat{g}(\beta) - \widehat{J}_n(\theta - \beta). \tag{C.32}$$

Local identification. Let $g_n(\theta) := \mathbb{E}\{h_i(y_i - x_i^\top \theta)\}$ denote the population combined projected moment along the local sequence. By (3.6),

$$g_n(\theta) = \frac{\tau}{\sqrt{n}} - J(\theta - \beta).$$

Hence the population GMM criterion is $g_n(\theta)^\top M g_n(\theta)$, whose first-order condition is

$$J^\top M \left\{ \frac{\tau}{\sqrt{n}} - J(\theta - \beta) \right\} = 0_p.$$

Since Assumption 4 (i) gives $J^\top M J \in \mathbb{S}^p_{++}$, the criterion is strictly convex in θ and has the unique minimizer

$$\beta_{M,n} = \beta + \frac{1}{\sqrt{n}}(J^\top M J)^{-1} J^\top M \tau = \beta + \frac{1}{\sqrt{n}} K_M \tau.$$

Thus the population GMM target is uniquely identified along the local sequence and satisfies $\beta_{M,n} \to \beta$.

We first establish the probability limit of the sample Jacobian. For the IV block,

$$\frac{1}{n} X^\top P_Z X = \left(\frac{1}{n} X^\top Z\right) \left(\frac{1}{n} Z^\top Z\right)^{-1} \left(\frac{1}{n} Z^\top X\right).$$

By the weak law of large numbers, $\frac{1}{n} X^\top Z \xrightarrow{\mathrm{p}} \Sigma_{xz}$, $\frac{1}{n} Z^\top Z \xrightarrow{\mathrm{p}} \Sigma_z$, and $\frac{1}{n} Z^\top X \xrightarrow{\mathrm{p}} \Sigma_{zx}$. Since $\Sigma_z \in \mathbb{S}^q_{++}$, the continuous mapping theorem gives

$$\frac{1}{n} X^\top P_Z X \xrightarrow{\mathrm{p}} \Sigma_{xz} \Sigma_z^{-1} \Sigma_{zx} = Q_{\mathrm{IV}}.$$

For the SV block,

$$\frac{1}{n} X^\top P_W^\perp X = \frac{1}{n} X^\top X - \left(\frac{1}{n} X^\top W\right) \left(\frac{1}{n} W^\top W\right)^{-1} \left(\frac{1}{n} W^\top X\right).$$

The weak law of large numbers gives $\frac{1}{n} X^\top X \xrightarrow{\mathrm{p}} \mathbb{E}(x_i x_i^\top)$, $\frac{1}{n} X^\top W \xrightarrow{\mathrm{p}} \Sigma_{xw}$, $\frac{1}{n} W^\top X \xrightarrow{\mathrm{p}} \Sigma_{wx}$, and $\frac{1}{n} W^\top W \xrightarrow{\mathrm{p}} \Sigma_w$. Since $\Sigma_w \in \mathbb{S}^m_{++}$,

$$\frac{1}{n} X^\top P_W^\perp X \xrightarrow{\mathrm{p}} \mathbb{E}(x_i x_i^\top) - \Sigma_{xw} \Sigma_w^{-1} \Sigma_{wx} = Q_{\mathrm{SV}}.$$

Hence,

$$\widehat{J}_n \xrightarrow{\mathrm{p}} \begin{pmatrix} Q_{\mathrm{IV}} \\ \\ Q_{\mathrm{SV}} \end{pmatrix} = J. \tag{C.33}$$

By Assumption 4 (i), $J^\top M J \in \mathbb{S}^p_{++}$. Thus, by (C.33) and the continuous mapping theorem,

$$\left(\widehat{J}_n^\top M \widehat{J}_n\right)^{-1} \widehat{J}_n^\top M \xrightarrow{\mathrm{p}} (J^\top M J)^{-1} J^\top M = K_M. \tag{C.34}$$

The first-order condition for minimizing $\widehat{g}(\theta)^\top M \widehat{g}(\theta)$ is $\widehat{J}_n^\top M \widehat{g}\{\widehat{\beta}(M)\} = 0$. Substituting (C.32) therefore gives, with probability approaching one,

$$\widehat{\beta}(M) - \beta = \left(\widehat{J}_n^\top M \widehat{J}_n\right)^{-1} \widehat{J}_n^\top M \widehat{g}(\beta). \tag{C.35}$$

Consistency. Under the local misspecification sequence,

$$\mathbb{E}(h_i \varsigma_i) = \frac{\tau}{\sqrt{n}} = \frac{1}{\sqrt{n}} \begin{pmatrix} \tau_z \\ \\ \tau_w \end{pmatrix}.$$

Thus, $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = \tau_z/\sqrt{n} \to 0_p$ and $\mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} = \tau_w/\sqrt{n} \to 0_p$.

For the IV block, since $Y - X\beta = \varsigma$,

$$\frac{1}{n} X^\top P_Z \varsigma = \left(\frac{1}{n} X^\top Z\right) \left(\frac{1}{n} Z^\top Z\right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n z_i \varsigma_i\right).$$

The weak law of large numbers along the local sequence and the continuous mapping theorem give

$$\frac{1}{n} X^\top P_Z \varsigma - \Sigma_{xz} \Sigma_z^{-1} \mathbb{E}(z_i \varsigma_i) \xrightarrow{\mathrm{p}} 0_p.$$

Since $\Sigma_{xz} \Sigma_z^{-1} \mathbb{E}(z_i \varsigma_i) = \mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = \tau_z/\sqrt{n} \to 0_p$, it follows that $\frac{1}{n} X^\top P_Z (Y - X\beta) \xrightarrow{\mathrm{p}} 0_p$.

For the SV block,

$$\frac{1}{n} X^\top P_W^\perp \varsigma = \frac{1}{n} X^\top \varsigma - \left(\frac{1}{n} X^\top W\right) \left(\frac{1}{n} W^\top W\right)^{-1} \left(\frac{1}{n} W^\top \varsigma\right).$$

Again by the weak law of large numbers and the continuous mapping theorem,

$$\frac{1}{n} X^\top P_W^\perp \varsigma - \left\{ \mathbb{E}(x_i \varsigma_i) - \Sigma_{xw} \Sigma_w^{-1} \mathbb{E}(w_i \varsigma_i) \right\} \xrightarrow{\text{p}} 0_p.$$

The term in braces equals $\mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} = \tau_w / \sqrt{n} \to 0_p$. Therefore, $\frac{1}{n} X^\top P_W^\perp (Y - X\beta) \xrightarrow{\text{p}} 0_p$.

Combining the two blocks gives

$$\widehat{g}(\beta) \xrightarrow{\text{p}} 0_{2p}. \tag{C.36}$$

Equations (C.35), (C.34), and (C.36) then imply

$$\widehat{\beta}(M) \xrightarrow{\text{p}} \beta.$$

Asymptotic normality. Suppose further that Assumption 4 (ii)–(iii) hold. Multiplying (C.35) by $\sqrt{n}$ gives

$$\sqrt{n}\{\widehat{\beta}(M) - \beta\} = \left(\widehat{J}_n^\top M \widehat{J}_n \right)^{-1} \widehat{J}_n^\top M \sqrt{n}\, \widehat{g}(\beta). \tag{C.37}$$

By (C.34), the matrix multiplying $\sqrt{n}\, \widehat{g}(\beta)$ converges in probability to K_M. It therefore remains to derive the limiting distribution of the normalized stacked sample moment.

We first obtain a first-order linear representation for its two projected moment blocks. For the IV block,

$$\frac{1}{\sqrt{n}} X^\top P_Z \varsigma = \left(\frac{1}{n} X^\top Z \right) \left(\frac{1}{n} Z^\top Z \right)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n z_i \varsigma_i.$$

Because $\mathbb{E}(z_i \varsigma_i)$ need not vanish, write $\varsigma_i = \{\varsigma_i - z_i^\top \Sigma_z^{-1} \mathbb{E}(z_i \varsigma_i)\} + z_i^\top \Sigma_z^{-1} \mathbb{E}(z_i \varsigma_i)$. Using this decomposition together with the equivalent representation of $\psi_{\text{IV},i}$ in (3.5), a direct rearrangement

gives

$$\frac{1}{\sqrt{n}}X^\top P_Z\varsigma - \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\text{IV},i}$$
$$= \left\{\left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1} - \Sigma_{xz}\Sigma_z^{-1}\right\}\frac{1}{\sqrt{n}}\sum_{i=1}^n z_i\left\{\varsigma_i - z_i^\top\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)\right\}.$$

The first factor is $o_p(1)$ by the same law-of-large-numbers and continuous-mapping argument used above. The summands in the second factor are centered because $\mathbb{E}[z_i\{\varsigma_i - z_i^\top\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)\}] = 0_q$. Moreover, Assumption 4 (ii) gives a uniformly bounded second moment for these summands. Hence their normalized sum is $O_p(1)$, and therefore

$$\frac{1}{\sqrt{n}}X^\top P_Z\varsigma = \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\text{IV},i} + o_p(1). \tag{C.38}$$

For the SV block, the same argument applies after removing the component of ς_i linearly explained by w_i. Since $P_W^\perp W = 0$,

$$P_W^\perp\varsigma = P_W^\perp\left\{\varsigma - W\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\right\}.$$

Expanding $P_W^\perp$ and comparing the resulting expression with $\psi_{\text{SV},i}$ gives

$$\frac{1}{\sqrt{n}}X^\top P_W^\perp\varsigma - \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\text{SV},i}$$
$$= -\left\{\left(\frac{1}{n}X^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1} - \Sigma_{xw}\Sigma_w^{-1}\right\}\frac{1}{\sqrt{n}}\sum_{i=1}^n w_i\left\{\varsigma_i - w_i^\top\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\right\}.$$

Again, the first factor is $o_p(1)$. The summands in the second factor are centered because $\mathbb{E}[w_i\{\varsigma_i - w_i^\top\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\}] = 0_m$, and their second moments are uniformly bounded by Assumption 4 (ii). Thus, the normalized sum is $O_p(1)$, so

$$\frac{1}{\sqrt{n}}X^\top P_W^\perp\varsigma = \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\text{SV},i} + o_p(1). \tag{C.39}$$

Stacking (C.38) and (C.39) yields the key first-order representation

$$\sqrt{n}\,\widehat{g}(\beta) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_i + o_p(1). \tag{C.40}$$

We next verify the mean and second-moment properties of the score. By the population IV projection property, $\mathbb{E}[\{x_i - \Pi_z x_i\} z_i^\top] = 0_{p\times q}$, and hence $\mathbb{E}(\psi_{\text{IV},i}) = \mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = \tau_z/\sqrt{n}$. Similarly, $\mathbb{E}\{(\Pi_w^\perp x_i) w_i^\top\} = 0_{p\times m}$, so $\mathbb{E}(\psi_{\text{SV},i}) = \mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} = \tau_w/\sqrt{n}$. Therefore,

$$\mathbb{E}(\psi_i) = \frac{\tau}{\sqrt{n}}. \tag{C.41}$$

It remains to verify that $\Sigma = \mathrm{Var}(\psi_i)$ is well defined. For the IV block, the equivalent score representation gives

$$\mathbb{E}\|\psi_{\text{IV},i}\|_2^2 \le 2\|\Sigma_{xz}\Sigma_z^{-1}\|_{\text{op}}^2 \mathbb{E}\left\| z_i \left\{ \varsigma_i - z_i^\top \Sigma_z^{-1} \mathbb{E}(z_i\varsigma_i) \right\} \right\|_2^2 + 2\left\|\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)\right\|_2^2 \mathbb{E}\|x_i z_i^\top\|_{\text{op}}^2.$$

The first term is finite by Assumption 4 (ii). For the second term, Cauchy–Schwarz gives $\mathbb{E}\|z_i\varsigma_i\|_2 \le \{\mathbb{E}\|z_i\|_2^2\}^{1/2}\{\mathbb{E}(\varsigma_i^2)\}^{1/2} < \infty$. Since $\Sigma_z \in \mathbb{S}_{++}^q$, $\|\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)\|_2 < \infty$, while $\mathbb{E}\|x_i z_i^\top\|_{\text{op}}^2 < \infty$ by Assumption 4 (ii). Consequently, $\mathbb{E}\|\psi_{\text{IV},i}\|_2^2 < \infty$.

For the SV block, first note that

$$\begin{aligned} &\mathbb{E}\left\| x_i \left\{ \varsigma_i - w_i^\top \Sigma_w^{-1} \mathbb{E}(w_i\varsigma_i) \right\} \right\|_2^2 \\ &\le 2\mathbb{E}\|x_i\varsigma_i\|_2^2 + 2\left\|\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\right\|_2^2 \mathbb{E}\|x_i w_i^\top\|_{\text{op}}^2. \end{aligned}$$

The first term is finite by Assumption 4 (ii). Moreover, Cauchy–Schwarz gives $\mathbb{E}\|w_i\varsigma_i\|_2 \le \{\mathbb{E}\|w_i\|_2^2\}^{1/2}\{\mathbb{E}(\varsigma_i^2)\}^{1/2} < \infty$. Since $\Sigma_w \in \mathbb{S}_{++}^m$, $\|\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\|_2 < \infty$, and $\mathbb{E}\|x_i w_i^\top\|_{\text{op}}^2 < \infty$ by Assumption 4 (ii). Hence the preceding expectation is finite.

Using

$$\psi_{\mathrm{SV},i} = x_i \left\{ \varsigma_i - w_i^\top \Sigma_w^{-1} \mathbb{E}(w_i \varsigma_i) \right\} - \Sigma_{xw} \Sigma_w^{-1} w_i \left\{ \varsigma_i - w_i^\top \Sigma_w^{-1} \mathbb{E}(w_i \varsigma_i) \right\},$$

we therefore obtain

$$\mathbb{E}\|\psi_{\mathrm{SV},i}\|_2^2 \le 2\mathbb{E} \left\| x_i \left\{ \varsigma_i - w_i^\top \Sigma_w^{-1} \mathbb{E}(w_i \varsigma_i) \right\} \right\|_2^2 + 2\|\Sigma_{xw} \Sigma_w^{-1}\|_{\mathrm{op}}^2 \mathbb{E} \left\| w_i \left\{ \varsigma_i - w_i^\top \Sigma_w^{-1} \mathbb{E}(w_i \varsigma_i) \right\} \right\|_2^2 < \infty,$$

where the last term is finite by Assumption 4 (ii). Thus, $\mathbb{E}\|\psi_i\|_2^2 < \infty$, and $\Sigma = \mathrm{Var}(\psi_i)$ is well defined along the local sequence.

We can now derive the limiting distribution of the normalized stacked score. By (C.41),

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_i = \frac{1}{\sqrt{n}} \sum_{i=1}^n \{\psi_i - \mathbb{E}(\psi_i)\} + \tau. \tag{C.42}$$

Thus, it is enough to establish a central limit theorem for the centered normalized sum.

Fix any $a \in \mathbb{R}^{2p}$. Along the local sequence, the scalar summands $a^\top \{\psi_i - \mathbb{E}(\psi_i)\}$ are row-wise independent and centered. Moreover,

$$\mathrm{Var}\left[a^\top \frac{1}{\sqrt{n}} \sum_{i=1}^n \{\psi_i - \mathbb{E}(\psi_i)\} \right] = a^\top \Sigma a \longrightarrow a^\top \Sigma a,$$

where the convergence follows from Assumption 4 (iii).

The same assumption also imposes the vector Lindeberg condition. Since $|a^\top v| \le \|a\|_2 \|v\|_2$ for every $v \in \mathbb{R}^{2p}$, it implies the corresponding scalar Lindeberg condition for $a^\top \{\psi_i - \mathbb{E}(\psi_i)\}$. The scalar Lindeberg–Feller central limit theorem therefore gives

$$a^\top \frac{1}{\sqrt{n}} \sum_{i=1}^n \{\psi_i - \mathbb{E}(\psi_i)\} \xrightarrow{\mathrm{d}} \mathcal{N}\left(0,\, a^\top \Sigma a\right).$$

Because this holds for every fixed $a \in \mathbb{R}^{2p}$, the Cramér–Wold device yields

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \{\psi_i - \mathbb{E}(\psi_i)\} \xrightarrow{\mathrm{d}} \mathcal{N}(0, \Sigma). \tag{C.43}$$

Combining (C.43) with (C.42), and recalling that τ is fixed along the local sequence, gives

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_i \xrightarrow{\mathrm{d}} \mathcal{N}(\tau, \Sigma). \tag{C.44}$$

The mean shift from 0 to τ is precisely the first-order effect of the $n^{-1/2}$-local moment violation.

The first-order representation (C.40) gives $\sqrt{n}\,\widehat{g}(\beta) = n^{-1/2} \sum_{i=1}^{n} \psi_i + o_p(1)$. Hence, combining this representation with (C.44) and applying Slutsky's theorem yields

$$\sqrt{n}\,\widehat{g}(\beta) \xrightarrow{\mathrm{d}} \mathcal{N}(\tau, \Sigma). \tag{C.45}$$

Finally, recall from (C.37) that

$$\sqrt{n}\{\widehat{\beta}(M) - \beta\} = \left(\widehat{J}_n^\top M \widehat{J}_n\right)^{-1} \widehat{J}_n^\top M \sqrt{n}\,\widehat{g}(\beta).$$

By (C.34), $(\widehat{J}_n^\top M \widehat{J}_n)^{-1} \widehat{J}_n^\top M \xrightarrow{\mathrm{p}} K_M$, while (C.45) gives $\sqrt{n}\,\widehat{g}(\beta) \xrightarrow{\mathrm{d}} \mathcal{N}(\tau, \Sigma)$. Another application of Slutsky's theorem therefore gives

$$\sqrt{n}\{\widehat{\beta}(M) - \beta\} \xrightarrow{\mathrm{d}} \mathcal{N}\left(K_M \tau,\, K_M \Sigma K_M^\top\right) = \mathcal{N}\{K_M \tau, V_{\mathrm{miss}}(M)\}. \tag{C.46}$$

Assumption 4 (iii) ensures that $V_{\mathrm{miss}}(M) \in \mathbb{S}_{++}^{p}$, so the limiting distribution is nondegenerate.

For the IV-only projected moment system, the same argument applies with $J = Q_{\mathrm{IV}}$, $\Sigma = \Sigma_{\mathrm{IV,miss}}$, $\tau = \tau_z$, and $M = M_z$. Likewise, for the SV-only system, it applies with $J = Q_{\mathrm{SV}}$,

$\Sigma = \Sigma_{\mathrm{SV,miss}}$, $\tau = \tau_w$, and $M = M_w$. Whenever the corresponding single-source Jacobian is nonsingular, $K_{M_z} = Q_{\mathrm{IV}}^{-1}$ for the IV-only system and $K_{M_w} = Q_{\mathrm{SV}}^{-1}$ for the SV-only system. Hence, in the exactly identified single-source systems, the local asymptotic bias and variance do not depend on the particular admissible weighting matrix.

This completes the proof.

□

C.6 Proof of Lemma 6

Proof of Lemma 6. For any fixed $\tau \in \mathcal{C}(D)$, Theorem 5 gives

$$\sqrt{n}\{\widehat{\beta}(M) - \beta\} \xrightarrow{\mathrm{d}} \mathcal{N}\left\{K_M\tau,\, K_M\Sigma K_M^\top\right\}.$$

By uniform integrability of the squared root-n estimation errors, the corresponding second moments converge. Hence

$$\mathrm{AMSE}_\tau(M) = \|K_M\tau\|_2^2 + \mathrm{tr}\left(K_M\Sigma K_M^\top\right),$$

using the identity $\mathbb{E}\|Z\|_2^2 = \|\mathbb{E}Z\|_2^2 + \mathrm{tr}\{\mathrm{Var}(Z)\}$.

Since the first-order risk depends on the weighting matrix only through its sensitivity, write K for any feasible sensitivity satisfying $KJ = I_p$. For $\mathcal{C}(D) = \{\tau = Du : \|u\|_2 \le 1\}$,

$$\begin{aligned} R(K) &= \sup_{\tau\in\mathcal{C}(D)} \left\{\|K\tau\|_2^2 + \mathrm{tr}(K\Sigma K^\top)\right\} \\ &= \sup_{\|u\|_2\le 1} \|KDu\|_2^2 + \mathrm{tr}(K\Sigma K^\top). \end{aligned}$$

By the definition of the operator norm,

$$\sup_{\|u\|_2 \leq 1} \|KDu\|_2^2 = \|KD\|_{\text{op}}^2.$$

Therefore,

$$R(K) = \|KD\|_{\text{op}}^2 + \text{tr}(K\Sigma K^\top),$$

which proves the result.

□

C.7 Proof of Proposition 7

Proof of Proposition 7. Under exact specification, the asymptotic squared-error risk associated with a sensitivity matrix K is $\text{tr}(K\Sigma K^\top)$, subject to the constraint $KJ = I_p$. Since $\Sigma \in \mathbb{S}_{++}^{2p}$ and $\text{rank}(J) = p$, the unique minimizer is

$$K_{\text{opt}} = (J^\top \Sigma^{-1} J)^{-1} J^\top \Sigma^{-1}.$$

Therefore,

$$\text{tr}(K_{\text{opt}} \Sigma K_{\text{opt}}^\top) = \text{tr}\left\{(J^\top \Sigma^{-1} J)^{-1}\right\}.$$

Because Q_{IV} and Q_{SV} are nonsingular, the IV-only and SV-only sensitivities are

$$K^{(\text{IV})} = \begin{pmatrix} Q_{\text{IV}}^{-1} & 0 \end{pmatrix}, \qquad K^{(\text{SV})} = \begin{pmatrix} 0 & Q_{\text{SV}}^{-1} \end{pmatrix}.$$

Both are feasible for the combined problem, since

$$K^{(\text{IV})} J = Q_{\text{IV}}^{-1} Q_{\text{IV}} = I_p, \qquad K^{(\text{SV})} J = Q_{\text{SV}}^{-1} Q_{\text{SV}} = I_p.$$

Hence, by optimality of K_{opt},

$$\text{tr}(K_{\text{opt}}\Sigma K_{\text{opt}}^\top) \leq \text{tr}\{K^{(\text{IV})}\Sigma(K^{(\text{IV})})^\top\},$$

and

$$\text{tr}(K_{\text{opt}}\Sigma K_{\text{opt}}^\top) \leq \text{tr}\{K^{(\text{SV})}\Sigma(K^{(\text{SV})})^\top\}.$$

Using the block structure of Σ,

$$\text{tr}\{K^{(\text{IV})}\Sigma(K^{(\text{IV})})^\top\} = \text{tr}\left(Q_{\text{IV}}^{-1}\Sigma_{\text{IV}}Q_{\text{IV}}^{-T}\right),$$

while

$$\text{tr}\{K^{(\text{SV})}\Sigma(K^{(\text{SV})})^\top\} = \text{tr}\left(Q_{\text{SV}}^{-1}\Sigma_{\text{SV}}Q_{\text{SV}}^{-T}\right).$$

Combining these identities gives

$$\text{tr}\left\{(J^\top\Sigma^{-1}J)^{-1}\right\} \leq \min\left[\text{tr}\left(Q_{\text{IV}}^{-1}\Sigma_{\text{IV}}Q_{\text{IV}}^{-T}\right), \text{tr}\left(Q_{\text{SV}}^{-1}\Sigma_{\text{SV}}Q_{\text{SV}}^{-T}\right)\right],$$

which proves the result.

□

C.8 Proof of Proposition 8

Proof of Proposition 8. By Lemma 6, $R(K)$ is the worst-case local AMSE associated with any feasible sensitivity K. By (4.3), K_{opt} minimizes $R(K)$ over the class $\{K : KJ = I_p\}$.

Since Q_{IV} and Q_{SV} are positive definite, the single-source sensitivities $K^{(\text{IV})} = (Q_{\text{IV}}^{-1}, 0)$ and $K^{(\text{SV})} = (0, Q_{\text{SV}}^{-1})$ are well defined. Moreover, using $J = (Q_{\text{IV}}^\top, Q_{\text{SV}}^\top)^\top$,

$$K^{(\text{IV})}J = Q_{\text{IV}}^{-1}Q_{\text{IV}} = I_p, \qquad K^{(\text{SV})}J = Q_{\text{SV}}^{-1}Q_{\text{SV}} = I_p.$$

Thus both single-source sensitivities are feasible points of (4.3).

Optimality of K_{opt} therefore gives

$$R(K_{\text{opt}}) \le R\{K^{(\text{IV})}\}, \qquad R(K_{\text{opt}}) \le R\{K^{(\text{SV})}\},$$

which proves the result.

□

C.9 Proof of Proposition 9

Proof of Proposition 9. Let $r(\beta) := Y - X\beta$. Recall that the stacked sample moment is

$$\widehat{g}(\beta) = \frac{1}{n}\begin{pmatrix} X^\top P_Z r(\beta) \\ \\ X^\top P_W^\perp r(\beta) \end{pmatrix}.$$

Under the block-diagonal weighting matrix M_λ, the restricted GMM criterion can be written as

$$\widehat{g}(\beta)^\top M_\lambda \widehat{g}(\beta) = \mathcal{J}_{Z,\lambda}(\beta) + \mathcal{J}_{W,\lambda}(\beta),$$

where

$$\mathcal{J}_{Z,\lambda}(\beta) := \frac{1-\lambda}{n} r(\beta)^\top P_Z X (X^\top P_Z X)^{-1} X^\top P_Z r(\beta),$$

$$\mathcal{J}_{W,\lambda}(\beta) := \frac{\lambda}{n} r(\beta)^\top P_W^\perp X (X^\top P_W^\perp X)^{-1} X^\top P_W^\perp r(\beta).$$

Indeed, for the projected IV block,

$$\left\{\frac{1}{n}X^\top P_Z r(\beta)\right\}^\top (1-\lambda)\left(\frac{1}{n}X^\top P_Z X\right)^{-1}\left\{\frac{1}{n}X^\top P_Z r(\beta)\right\}$$

$$= \frac{1-\lambda}{n} r(\beta)^\top P_Z X (X^\top P_Z X)^{-1} X^\top P_Z r(\beta),$$

and the same calculation gives the SV component.

Since $\partial r(\beta)/\partial \beta^\top = -X$, and $P_Z X (X^\top P_Z X)^{-1} X^\top P_Z$ is symmetric, differentiation of the IV component gives

$$\begin{aligned}\nabla_\beta \mathcal{J}_{Z,\lambda}(\beta) &= -\frac{2(1-\lambda)}{n} X^\top P_Z X (X^\top P_Z X)^{-1} X^\top P_Z r(\beta) \\ &= -\frac{2(1-\lambda)}{n} X^\top P_Z r(\beta).\end{aligned}$$

Similarly,

$$\begin{aligned}\nabla_\beta \mathcal{J}_{W,\lambda}(\beta) &= -\frac{2\lambda}{n} X^\top P_W^\perp X (X^\top P_W^\perp X)^{-1} X^\top P_W^\perp r(\beta) \\ &= -\frac{2\lambda}{n} X^\top P_W^\perp r(\beta).\end{aligned}$$

Therefore, the first-order condition for minimizing $\widehat{g}(\beta)^\top M_\lambda \widehat{g}(\beta)$ is

$$(1-\lambda) X^\top P_Z r(\beta) + \lambda X^\top P_W^\perp r(\beta) = 0.$$

Substituting $r(\beta) = Y - X\beta$ gives

$$\begin{aligned}&(1-\lambda) X^\top P_Z Y + \lambda X^\top P_W^\perp Y \\ &\quad - \Big\{(1-\lambda) X^\top P_Z X + \lambda X^\top P_W^\perp X\Big\}\beta = 0.\end{aligned}$$

Hence, whenever the weighted Gram matrix is nonsingular, the minimizer is

$$\widehat{\beta}_\lambda = \Big\{(1-\lambda) X^\top P_Z X + \lambda X^\top P_W^\perp X\Big\}^{-1} \Big\{(1-\lambda) X^\top P_Z Y + \lambda X^\top P_W^\perp Y\Big\},$$

which is precisely the weighted least-squares estimator defined in Section 4.3.

For $\lambda \in (0,1)$, both diagonal blocks of M_λ are positive definite, so $M_\lambda \in \mathbb{S}^{2p}_{++}$ and the preceding calculation gives the stated restricted-GMM representation within the weighting class of Section 3.1. At $\lambda = 0$ and $\lambda = 1$, M_λ is positive semidefinite rather than positive definite,

but the same quadratic criterion remains well defined and the preceding first-order calculation yields the IV-only and SV-only endpoint estimators, respectively.

□

C.10 Proof of Proposition 10

Proof of Proposition 10. Let

$$\mathcal{D} := \sigma(X, Z, W)$$

denote the sigma-field generated by the realized design matrices. Throughout the proof, all conditional expectations, variances, and covariances are taken with respect to $\mathcal{D}$.

For notational convenience, define the two score blocks

$$U_Z := X^\top P_Z \varsigma, \qquad U_W := X^\top P_W^\perp \varsigma,$$

and their weighted combination

$$U_\lambda := (1-\lambda)U_Z + \lambda U_W.$$

Using $Y = X\beta + \varsigma$ in (4.4), we obtain

$$\begin{aligned}
\widehat{\beta}_\lambda &= S_\lambda^{-1}\left\{(1-\lambda)X^\top P_Z Y + \lambda X^\top P_W^\perp Y\right\} \\
&= S_\lambda^{-1}\left[(1-\lambda)X^\top P_Z(X\beta+\varsigma) + \lambda X^\top P_W^\perp(X\beta+\varsigma)\right] \\
&= S_\lambda^{-1}\left[\left\{(1-\lambda)X^\top P_Z X + \lambda X^\top P_W^\perp X\right\}\beta + U_\lambda\right].
\end{aligned}$$

By the definition

$$S_\lambda = (1-\lambda)S_Z + \lambda S_W = (1-\lambda)X^\top P_Z X + \lambda X^\top P_W^\perp X,$$

the preceding display reduces to

$$\widehat{\beta}_\lambda - \beta = S_\lambda^{-1} U_\lambda = S_\lambda^{-1} \left\{ (1-\lambda) X^\top P_Z \varsigma + \lambda X^\top P_W^\perp \varsigma \right\}.$$

Define

$$\mu_Z := \mathbb{E}(U_Z \mid \mathcal{D}), \qquad \mu_W := \mathbb{E}(U_W \mid \mathcal{D}),$$

and hence

$$\mu_\lambda := \mathbb{E}(U_\lambda \mid \mathcal{D}) = (1-\lambda)\mu_Z + \lambda \mu_W.$$

Because S_λ is measurable with respect to $\mathcal{D}$,

$$\mathbb{E}(\widehat{\beta}_\lambda - \beta \mid \mathcal{D}) = S_\lambda^{-1} \mu_\lambda.$$

Thus, the estimator need not be conditionally centered even when the population moment restrictions are valid.

Because S_λ is symmetric positive definite, S_λ^{-1} is also symmetric. Therefore,

$$\begin{aligned} \mathcal{R}_{\mathrm{tr}}(\lambda) &= \mathbb{E}\Big(\|\widehat{\beta}_\lambda - \beta\|_2^2 \mid \mathcal{D} \Big) \\ &= \mathrm{tr}\left[S_\lambda^{-1} \mathbb{E}(U_\lambda U_\lambda^\top \mid \mathcal{D}) S_\lambda^{-1} \right]. \end{aligned}$$

Using

$$\mathbb{E}(U_\lambda U_\lambda^\top \mid \mathcal{D}) = \mathrm{Var}(U_\lambda \mid \mathcal{D}) + \mu_\lambda \mu_\lambda^\top,$$

we equivalently obtain

$$\mathcal{R}_{\mathrm{tr}}(\lambda) = \mathrm{tr}\left[S_\lambda^{-1} \mathrm{Var}(U_\lambda \mid \mathcal{D}) S_\lambda^{-1} \right] + \|S_\lambda^{-1} \mu_\lambda\|_2^2.$$

This proves the conditional bias–variance decomposition in (4.5).

Define

$$\Omega_\lambda := \mathbb{E}(U_\lambda U_\lambda^\top \mid \mathcal{D}).$$

Then $\Omega_\lambda \succeq 0$. By linearity,

$$\begin{aligned}\Omega_\lambda = (1-\lambda)^2\mathbb{E}(U_Z U_Z^\top \mid \mathcal{D}) \\ + \lambda(1-\lambda)\mathbb{E}(U_Z U_W^\top \mid \mathcal{D}) \\ + \lambda(1-\lambda)\mathbb{E}(U_W U_Z^\top \mid \mathcal{D}) \\ + \lambda^2\mathbb{E}(U_W U_W^\top \mid \mathcal{D}).\end{aligned}$$

Taking traces and using

$$\operatorname{tr}(U_Z U_W^\top) = U_Z^\top U_W$$

gives

$$\operatorname{tr}(\Omega_\lambda) = (1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W.$$

For later use, write

$$N(\lambda) := (1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W.$$

The cyclic property of the trace gives

$$\mathcal{R}_{\mathrm{tr}}(\lambda) = \operatorname{tr}\left(\Omega_\lambda S_\lambda^{-2}\right).$$

We use the following elementary trace inequality. For any two positive semidefinite matrices A and B of the same dimension,

$$\lambda_{\min}(B)\operatorname{tr}(A) \le \operatorname{tr}(AB) \le \lambda_{\max}(B)\operatorname{tr}(A).$$

Indeed,

$$\lambda_{\min}(B)I \preceq B \preceq \lambda_{\max}(B)I.$$

Premultiplying and postmultiplying by $A^{1/2}$ gives

$$\lambda_{\min}(B)A \preceq A^{1/2}BA^{1/2} \preceq \lambda_{\max}(B)A.$$

Taking traces and using $\operatorname{tr}(A^{1/2}BA^{1/2}) = \operatorname{tr}(AB)$ establishes the claimed inequality.

Applying this inequality with

$$A = \Omega_\lambda, \qquad B = S_\lambda^{-2},$$

yields

$$\lambda_{\min}(S_\lambda^{-2}) \operatorname{tr}(\Omega_\lambda) \le \mathcal{R}_{\mathrm{tr}}(\lambda) \le \lambda_{\max}(S_\lambda^{-2}) \operatorname{tr}(\Omega_\lambda).$$

If $s_1, \dots, s_p$ are the eigenvalues of S_λ, then the eigenvalues of S_λ^{-2} are $s_1^{-2}, \dots, s_p^{-2}$. Therefore,

$$\lambda_{\min}(S_\lambda^{-2}) = \lambda_{\max}(S_\lambda)^{-2}, \qquad \lambda_{\max}(S_\lambda^{-2}) = \lambda_{\min}(S_\lambda)^{-2}.$$

Since $\operatorname{tr}(\Omega_\lambda) = N(\lambda)$, we obtain

$$\frac{(1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W}{\lambda_{\max}(S_\lambda)^2} \le \mathcal{R}_{\mathrm{tr}}(\lambda) \le \frac{(1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W}{\lambda_{\min}(S_\lambda)^2}.$$

We now establish the uniform comparison with the surrogate risk. Define

$$a_Z := \lambda_{\min}(S_Z), \qquad a_W := \lambda_{\min}(S_W),$$

and

$$d_\lambda := (1-\lambda)a_Z + \lambda a_W.$$

Because $S_\lambda \succ 0$ for every $\lambda \in [0,1]$, in particular $S_Z = S_0 \succ 0$ and $S_W = S_1 \succ 0$. Hence

$$a_Z > 0, \qquad a_W > 0, \qquad d_\lambda > 0.$$

For every unit vector $v \in \mathbb{R}^p$,

$$\begin{aligned} v^\top S_\lambda v &= (1-\lambda) v^\top S_Z v + \lambda v^\top S_W v \\ &\geq (1-\lambda)\lambda_{\min}(S_Z) + \lambda \lambda_{\min}(S_W) \\ &= d_\lambda. \end{aligned}$$

Taking the minimum over all unit vectors gives

$$\lambda_{\min}(S_\lambda) \geq d_\lambda.$$

Similarly,

$$v^\top S_\lambda v \leq (1-\lambda)\lambda_{\max}(S_Z) + \lambda \lambda_{\max}(S_W),$$

and hence

$$\lambda_{\max}(S_\lambda) \leq (1-\lambda)\lambda_{\max}(S_Z) + \lambda \lambda_{\max}(S_W).$$

Under

$$\lambda_{\max}(S_Z) \lesssim \lambda_{\min}(S_Z), \qquad \lambda_{\max}(S_W) \lesssim \lambda_{\min}(S_W),$$

there exists a constant $C \geq 1$, independent of $\lambda \in [0,1]$, such that

$$\lambda_{\max}(S_Z) \leq C a_Z, \qquad \lambda_{\max}(S_W) \leq C a_W.$$

It follows that

$$\lambda_{\max}(S_\lambda) \leq C\{(1-\lambda)a_Z + \lambda a_W\} = C d_\lambda.$$

Combining the preceding bounds gives

$$d_\lambda \le \lambda_{\min}(S_\lambda) \le \lambda_{\max}(S_\lambda) \le C d_\lambda$$

uniformly over $\lambda \in [0,1]$.

Substituting these inequalities into the trace-risk bounds yields

$$\frac{N(\lambda)}{C^2 d_\lambda^2} \le \mathcal{R}_{\mathrm{tr}}(\lambda) \le \frac{N(\lambda)}{d_\lambda^2}.$$

By definition,

$$\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) = \frac{N(\lambda)}{d_\lambda^2} = \frac{(1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W}{\{(1-\lambda)a_Z + \lambda a_W\}^2}.$$

Therefore,

$$C^{-2}\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) \le \mathcal{R}_{\mathrm{tr}}(\lambda) \le \widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda)$$

uniformly over $\lambda \in [0,1]$, which proves

$$\mathcal{R}_{\mathrm{tr}}(\lambda) \asymp \widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) \qquad \text{uniformly over } \lambda \in [0,1].$$

It remains to characterize the stationary points of the surrogate risk. Retain the definition of $N(\lambda)$ above and let

$$D(\lambda) := (1-\lambda)a_Z + \lambda a_W.$$

Then

$$\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) = \frac{N(\lambda)}{D(\lambda)^2}.$$

Expanding $N(\lambda)$ gives

$$N(\lambda) = c_Z + 2(c_{ZW} - c_Z)\lambda + (c_Z - 2c_{ZW} + c_W)\lambda^2,$$

so

$$N'(\lambda) = 2(c_{ZW} - c_Z) + 2(c_Z - 2c_{ZW} + c_W)\lambda.$$

Moreover,

$$D(\lambda) = a_Z + (a_W - a_Z)\lambda, \qquad D'(\lambda) = a_W - a_Z.$$

Since $D(\lambda) > 0$ on $[0, 1]$, differentiation gives

$$\widetilde{\mathcal{R}}'_{\mathrm{tr}}(\lambda) = \frac{N'(\lambda)D(\lambda) - 2N(\lambda)D'(\lambda)}{D(\lambda)^3}.$$

Thus, every interior stationary point must satisfy

$$N'(\lambda)D(\lambda) - 2N(\lambda)D'(\lambda) = 0.$$

A direct expansion shows that the quadratic terms cancel and that

$$\frac{1}{2}\left\{N'(\lambda)D(\lambda) - 2N(\lambda)D'(\lambda)\right\} = \Big\{(c_W - c_{ZW})a_Z + (c_Z - c_{ZW})a_W\Big\}\lambda$$

$$- \left(c_Z a_W - c_{ZW} a_Z\right).$$

Therefore, provided

$$(c_W - c_{ZW})a_Z + (c_Z - c_{ZW})a_W \neq 0,$$

the stationary-point equation is linear and has the unique solution

$$\lambda^\star_{\mathrm{sur}} = \frac{c_Z a_W - c_{ZW} a_Z}{(c_W - c_{ZW})a_Z + (c_Z - c_{ZW})a_W}.$$

Consequently, there is at most one interior stationary point. Substituting back

$$a_Z = \lambda_{\min}(S_Z), \qquad a_W = \lambda_{\min}(S_W)$$

gives the expression stated in the proposition.

Finally, $\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda)$ is continuous on the compact interval $[0,1]$, because $D(\lambda) > 0$ throughout this interval. Hence it attains a global minimum. Any minimizer must occur either at a boundary point or at an interior stationary point. It is therefore sufficient to compare

$$\widetilde{\mathcal{R}}_{\mathrm{tr}}(0), \qquad \widetilde{\mathcal{R}}_{\mathrm{tr}}(1),$$

and, whenever $\lambda^{\star}_{\mathrm{sur}} \in (0,1)$, the additional candidate

$$\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda^{\star}_{\mathrm{sur}}).$$

This completes the proof. □

D. Projection geometry under low-rank factor structure

Let $U \in \mathbb{R}^{n\times r}$ and $\Gamma \in \mathbb{R}^{q\times r}$, and write $B := \Gamma^\top$ so that the (rank-r) signal is

$$S := U\Gamma^\top = UB \in \mathbb{R}^{n\times q}.$$

For any full-column-rank matrix A, define the orthogonal projector onto its column space by

$$P_A := A(A^\top A)^{-1}A^\top \in \mathbb{R}^{n\times n}.$$

Noiseless case: $W = U\Gamma^\top$. Since each column of $W = UB$ is a linear combination of columns of U, we have $\mathcal{C}(W) \subseteq \mathcal{C}(U)$, and therefore the corresponding orthogonal projections satisfy

$$P_U P_W = P_W, \qquad P_W P_U = P_W, \qquad \text{and} \qquad P_U - P_W \succeq 0.$$

Moreover, $P_W = P_U$ holds if and only if $\mathcal{C}(W) = \mathcal{C}(U)$, equivalently $\operatorname{rank}(UB) = \operatorname{rank}(U) = r$; in particular, if U and Γ are both full column rank r, then $\mathcal{C}(W) = \mathcal{C}(U)$ and $P_W = P_U$.

Noisy case: $W = U\Gamma^\top + E$. Exact nesting is retained only when the noise does not inject components outside $\mathcal{C}(U)$:

$$\mathcal{C}(W) \subseteq \mathcal{C}(U) \quad \Longleftrightarrow \quad (I - P_U)W = 0 \quad \Longleftrightarrow \quad (I - P_U)E = 0.$$

Hence, even when U and Γ are full column rank, generic noise E destroys the identities $P_U P_W = P_W$ and $P_W = P_U$.

A stable analogue is obtained by projecting onto the leading rank-r left singular subspace of W. Let $\widehat{U}_r \in \mathbb{R}^{n\times r}$ collect the top r left singular vectors of W, and define $\widehat{P}_r := \widehat{U}_r\widehat{U}_r^\top$. If

$\|E\|_2 < \sigma_r(S)$ (where $\sigma_r(S)$ is the smallest nonzero singular value of S), then standard subspace perturbation bounds yield

$$\|\widehat{P}_r - P_U\|_2 = \|\widehat{P}_r - P_S\|_2 \;\le\; \frac{2\|E\|_2}{\sigma_r(S) - \|E\|_2} \;\lesssim\; \frac{\|E\|_2}{\sigma_r(S)}.$$

In addition, $\sigma_r(S) \ge \sigma_r(U)\sigma_r(\Gamma)$ provides a convenient lower bound, so subspace (and hence projector) consistency holds when $\|E\|_2 \ll \sigma_r(U)\sigma_r(\Gamma)$.

E. Raw versus projected IV information

The projected representation also places the IV and SV restrictions on the same p-dimensional scale. The next result shows that this compression does not sacrifice any IV identifying direction.

Proposition 11 (Projection preserves identifying directions)**.** Under (2.1), suppose the following conditions hold.

(i) The instrument covariance matrix satisfies $\Sigma_z \in \mathbb{S}^q_{++}$.

(ii) The projected IV and residualized SV moments are well defined, with $Q_{\text{IV}} = \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx}$ and $Q_{\text{SV}} = \mathbb{E}\{(\Pi_w^{\perp} x_i)x_i^{\top}\}$.

Then $\ker(Q_{\text{IV}}) = \ker(\Sigma_{zx})$. Consequently, for $J = \{Q_{\text{IV}}^{\top}, Q_{\text{SV}}^{\top}\}^{\top}$ and $J_{\text{raw}} := \{\Sigma_{zx}^{\top}, Q_{\text{SV}}^{\top}\}^{\top}$, we have $\ker(J) = \ker(J_{\text{raw}})$ and $\operatorname{rank}(J) = \operatorname{rank}(J_{\text{raw}})$. Hence, the projected-IV-plus-residualized-SV system and the corresponding raw-IV-plus-residualized-SV system preserve exactly the same identifying directions for β, and $\operatorname{rank}(J) = p$ if and only if $\operatorname{rank}(J_{\text{raw}}) = p$.

Moreover, if $\operatorname{rank}(\Sigma_{zx}) = p$, any linear compression of the q-dimensional IV moment that preserves all IV identifying directions must have dimension at least p. The projected instrument $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i$ has dimension p and therefore attains this minimum.

The main analysis adopts the projected IV moment because it places the IV and SV channels on a common p-dimensional scale while preserving all IV identifying directions. Projection, however, need not preserve all efficiency information contained in the original q-dimensional IV moment. This section isolates that distinction by comparing the raw and projected IV systems directly, without involving the SV block.

Recall that the projected IV restriction is $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$, whereas the classical raw-IV restriction is $\mathbb{E}(z_i\varsigma_i) = 0_q$. Since $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)$, classical IV validity implies projected IV validity. The converse need not hold when $q > p$, because components of $\mathbb{E}(z_i\varsigma_i)$ lying in $\ker(\Sigma_{xz}\Sigma_z^{-1})$ are removed by projection and therefore do not affect the projected restriction. Thus, the comparison between the two representations involves a validity–efficiency trade-off. The raw IV system retains all q moment restrictions and may exploit additional overidentifying information, but requires the stronger condition that all of these directions are valid. The projected system requires validity only after compression to the component relevant for predicting x_i, at the possible cost of discarding useful overidentifying information when the stronger classical restriction is in fact correct.

For the remainder of this section, suppose that $\mathbb{E}(z_i\varsigma_i) = 0_q$. Define the raw IV score by $\psi^{\mathrm{IV}}_{\mathrm{raw},i} := z_i\varsigma_i \in \mathbb{R}^q$, with $\Sigma^{\mathrm{IV}}_{\mathrm{raw}} := \operatorname{Var}(\psi^{\mathrm{IV}}_{\mathrm{raw},i})$, and define $T_{\mathrm{IV}} := \Sigma_{xz}\Sigma_z^{-1} \in \mathbb{R}^{p\times q}$. Under

classical IV validity, the projected IV score satisfies $\psi_{\text{IV},i} = T_{\text{IV}}\psi^{\text{IV}}_{\text{raw},i}$, so that $\Sigma_{\text{IV}} = T_{\text{IV}}\Sigma^{\text{IV}}_{\text{raw}}T_{\text{IV}}^{\top}$. The corresponding Jacobians are Σ_{zx} for the raw IV system and $Q_{\text{IV}} = T_{\text{IV}}\Sigma_{zx}$ for the projected system.

Let $\mathcal{I}^{\text{IV}}_{\text{raw}} := \Sigma_{zx}^{\top}(\Sigma^{\text{IV}}_{\text{raw}})^{-1}\Sigma_{zx}$ and $\mathcal{I}^{\text{IV}}_{\text{proj}} := Q_{\text{IV}}^{\top}\Sigma_{\text{IV}}^{-1}Q_{\text{IV}}$ denote the efficient information matrices in the raw and projected IV systems, respectively. For the equality characterization, define

$$P_{T_{\text{IV}}} := (\Sigma^{\text{IV}}_{\text{raw}})^{1/2}T_{\text{IV}}^{\top}(T_{\text{IV}}\Sigma^{\text{IV}}_{\text{raw}}T_{\text{IV}}^{\top})^{-1}T_{\text{IV}}(\Sigma^{\text{IV}}_{\text{raw}})^{1/2}$$

.

Proposition 12 (Raw IV information dominates projected IV information)**.** Suppose the following conditions hold.

(i) The classical IV restriction is correctly specified, $\mathbb{E}(z_i\varsigma_i) = 0_q$.

(ii) The projected IV block is individually identified, $Q_{\text{IV}} \in \mathbb{S}^p_{++}$, and the projection is well defined with $\Sigma_z \in \mathbb{S}^q_{++}$.

(iii) The regularity conditions required for the first-order expansions of the raw and projected IV systems hold, and $\Sigma^{\text{IV}}_{\text{raw}} \in \mathbb{S}^q_{++}$.

Then

$$\mathcal{I}^{\text{IV}}_{\text{proj}} \preceq \mathcal{I}^{\text{IV}}_{\text{raw}}, \qquad V^{\star}_{\text{raw,IV}} := (\mathcal{I}^{\text{IV}}_{\text{raw}})^{-1} \preceq (\mathcal{I}^{\text{IV}}_{\text{proj}})^{-1} =: V^{\star}_{\text{proj,IV}}. \tag{E.1}$$

More precisely,

$$\mathcal{I}^{\text{IV}}_{\text{raw}} - \mathcal{I}^{\text{IV}}_{\text{proj}} = \Sigma_{zx}^{\top}(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}(I_q - P_{T_{\text{IV}}})(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}\Sigma_{zx} \succeq 0. \tag{E.2}$$

Hence, projection is information-lossless if and only if $\mathcal{C}\{(\Sigma_{\text{raw}}^{\text{IV}})^{-1}\Sigma_{zx}\} \subseteq \mathcal{C}(T_{\text{IV}}^{\top})$.

Proposition 12 separates preservation of identification from preservation of efficiency. By Proposition 11, the projected IV moment preserves exactly the same identifying directions as the raw IV moment. Hence, the additional directions available when $q > p$ are not required for point identification of β. They may nevertheless contain useful overidentifying information. When the stronger classical restriction $\mathbb{E}(z_i\varsigma_i) = 0_q$ is valid, retaining these directions can reduce asymptotic variance, whereas projection may discard part of this information.

The equality condition has a direct geometric interpretation. The space $\mathcal{C}(T_{\text{IV}}^{\top})$ contains the raw-IV directions retained by the projection, whereas $\mathcal{C}\{(\Sigma_{\text{raw}}^{\text{IV}})^{-1}\Sigma_{zx}\}$ contains the efficient raw-IV directions relevant for estimating β. Projection is therefore information-lossless precisely when every efficient raw-IV direction is already contained in the subspace retained by T_{IV}. If some efficient direction lies outside that subspace, the projected system still preserves identification but loses information that could otherwise improve precision.

This result clarifies the trade-off underlying the projected formulation. The raw IV system can be more efficient when all q classical IV moment restrictions are valid, but this gain relies on the stronger validity requirement. The projected formulation instead requires only the weaker projected restriction in the main analysis and provides an identification-preserving p-dimensional representation that can be placed on the same scale as the surrogate-control moment. Its role is therefore not to dominate the raw IV system in efficiency, but to provide the common representation required for the unified IV–SV analysis.

When $q = p$ and $Q_{\text{IV}} \in \mathbb{S}^p_{++}$, T_{IV} is nonsingular, so the raw and projected IV moments are related by an invertible transformation. Consequently, $\mathcal{I}^{\text{IV}}_{\text{proj}} = \mathcal{I}^{\text{IV}}_{\text{raw}}$, and no IV information is lost through projection. The validity–efficiency trade-off is therefore relevant primarily in the overidentified case $q > p$.

E.1 Proof of Proposition 11

Proof of Proposition 11. We first show that the projected IV Jacobian and the raw IV Jacobian have the same null space. Recall that $Q_{\text{IV}} = \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx}$. For any $a \in \mathbb{R}^p$,

$$a^\top Q_{\text{IV}} a = a^\top \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx} a = \left\| \Sigma_z^{-1/2}\Sigma_{zx} a \right\|_2^2 .$$

Since $\Sigma_z \in \mathbb{S}^q_{++}$, the right-hand side is zero if and only if $\Sigma_{zx} a = 0_q$. Moreover, $Q_{\text{IV}} \succeq 0$, so $Q_{\text{IV}} a = 0_p$ if and only if $a^\top Q_{\text{IV}} a = 0$. Therefore,

$$\ker(Q_{\text{IV}}) = \ker(\Sigma_{zx}). \tag{E.3}$$

We next compare the identifying directions of the projected and raw combined systems. By definition,

$$J = \begin{pmatrix} Q_{\text{IV}} \\ \\ Q_{\text{SV}} \end{pmatrix}, \qquad J_{\text{raw}} = \begin{pmatrix} \Sigma_{zx} \\ \\ Q_{\text{SV}} \end{pmatrix}.$$

For any $a \in \mathbb{R}^p$, $Ja = 0_{2p}$ if and only if $Q_{\text{IV}} a = 0_p$ and $Q_{\text{SV}} a = 0_p$. By (E.3), this is equivalent to $\Sigma_{zx} a = 0_q$ and $Q_{\text{SV}} a = 0_p$, which in turn is equivalent to $J_{\text{raw}} a = 0_{q+p}$. Hence,

$$\ker(J) = \ker(J_{\text{raw}}).$$

Because both matrices have p columns, the rank–nullity theorem then gives

$$\operatorname{rank}(J) = \operatorname{rank}(J_{\text{raw}}).$$

Thus, replacing the raw IV block by the projected IV block does not remove any parameter direction that can be identified by the combined IV–SV system. In particular, $\operatorname{rank}(J) = p$ if and only if $\operatorname{rank}(J_{\text{raw}}) = p$.

It remains to establish the minimal dimension of an identification-preserving linear IV compression. Suppose $\operatorname{rank}(\Sigma_{zx}) = p$, and consider any k-dimensional linear compression of the raw q-dimensional IV moment. The Jacobian of the resulting compressed moment has rank at most k. Therefore, preserving all p identifying directions of Σ_{zx} necessarily requires $k \geq p$.

Finally, $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i \in \mathbb{R}^p$, so the projected IV moment has exactly p components. By (E.3), its Jacobian Q_{IV} has the same null space, and hence the same identifying directions, as Σ_{zx}. Therefore, the projected IV block preserves all IV identifying directions using the smallest possible number p of linearly compressed moments.

This completes the proof.

□

E.2 Proof of Proposition 12

Proof of Proposition 12. Recall that $T_{\text{IV}} = \Sigma_{xz}\Sigma_z^{-1}$, $\Sigma_{\text{IV}} = T_{\text{IV}}\Sigma_{\text{raw}}^{\text{IV}}T_{\text{IV}}^\top$, and $Q_{\text{IV}} = T_{\text{IV}}\Sigma_{zx}$. The corresponding efficient information matrices are $\mathcal{I}_{\text{raw}}^{\text{IV}} = \Sigma_{zx}^\top(\Sigma_{\text{raw}}^{\text{IV}})^{-1}\Sigma_{zx}$ and $\mathcal{I}_{\text{proj}}^{\text{IV}} = Q_{\text{IV}}^\top\Sigma_{\text{IV}}^{-1}Q_{\text{IV}}$.

We first verify that Σ_{IV} is nonsingular. Since $Q_{\text{IV}} = T_{\text{IV}}\Sigma_{zx} \in \mathbb{S}^p_{++}$, the matrix $T_{\text{IV}} \in \mathbb{R}^{p\times q}$ has full row rank p. Together with $\Sigma^{\text{IV}}_{\text{raw}} \in \mathbb{S}^q_{++}$, this implies $\Sigma_{\text{IV}} = T_{\text{IV}}\Sigma^{\text{IV}}_{\text{raw}}T_{\text{IV}}^\top \in \mathbb{S}^p_{++}$. Hence, Σ_{IV}^{-1} and $P_{T_{\text{IV}}}$ are well defined.

Using $Q_{\text{IV}} = T_{\text{IV}}\Sigma_{zx}$, we can write

$$\begin{aligned}\mathcal{I}^{\text{IV}}_{\text{proj}} &= \Sigma_{zx}^\top T_{\text{IV}}^\top \left(T_{\text{IV}}\Sigma^{\text{IV}}_{\text{raw}}T_{\text{IV}}^\top\right)^{-1} T_{\text{IV}}\Sigma_{zx} \\ &= \Sigma_{zx}^\top (\Sigma^{\text{IV}}_{\text{raw}})^{-1/2} P_{T_{\text{IV}}} (\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}\Sigma_{zx}, \end{aligned} \tag{E.4}$$

where $P_{T_{\text{IV}}} = (\Sigma^{\text{IV}}_{\text{raw}})^{1/2}T_{\text{IV}}^\top(T_{\text{IV}}\Sigma^{\text{IV}}_{\text{raw}}T_{\text{IV}}^\top)^{-1}T_{\text{IV}}(\Sigma^{\text{IV}}_{\text{raw}})^{1/2}$.

The matrix $P_{T_{\text{IV}}}$ is symmetric. Moreover, $P^2_{T_{\text{IV}}} = P_{T_{\text{IV}}}$, since the middle product $T_{\text{IV}}\Sigma^{\text{IV}}_{\text{raw}}T_{\text{IV}}^\top$ cancels with its inverse. Hence, $P_{T_{\text{IV}}}$ is the orthogonal projector onto $\mathcal{C}\{(\Sigma^{\text{IV}}_{\text{raw}})^{1/2}T_{\text{IV}}^\top\}$. It follows that $I_q - P_{T_{\text{IV}}} \succeq 0$.

The raw information matrix can similarly be written as $\mathcal{I}^{\text{IV}}_{\text{raw}} = \Sigma_{zx}^\top(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}I_q(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}\Sigma_{zx}$. Subtracting (E.4) therefore yields

$$\mathcal{I}^{\text{IV}}_{\text{raw}} - \mathcal{I}^{\text{IV}}_{\text{proj}} = \Sigma_{zx}^\top(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}(I_q - P_{T_{\text{IV}}})(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}\Sigma_{zx} \succeq 0. \tag{E.5}$$

Thus, $\mathcal{I}^{\text{IV}}_{\text{proj}} \preceq \mathcal{I}^{\text{IV}}_{\text{raw}}$.

We next translate the information ordering into an asymptotic covariance ordering. Since $Q_{\text{IV}} \in \mathbb{S}^p_{++}$, the projected Jacobian has full rank p. By $\ker(Q_{\text{IV}}) = \ker(\Sigma_{zx})$ from Proposition 11, Σ_{zx} also has full column rank p. Together with positive definiteness of $\Sigma^{\text{IV}}_{\text{raw}}$ and Σ_{IV}, this gives $\mathcal{I}^{\text{IV}}_{\text{raw}}, \mathcal{I}^{\text{IV}}_{\text{proj}} \in \mathbb{S}^p_{++}$. Since inversion reverses the Loewner order on positive definite matrices,

$$V^\star_{\text{raw,IV}} = (\mathcal{I}^{\text{IV}}_{\text{raw}})^{-1} \preceq (\mathcal{I}^{\text{IV}}_{\text{proj}})^{-1} = V^\star_{\text{proj,IV}}.$$

It remains to characterize when projection is information-lossless. By (E.5), $\mathcal{I}^{\text{IV}}_{\text{raw}} = \mathcal{I}^{\text{IV}}_{\text{proj}}$ if and only if

$$(I_q - P_{T_{\text{IV}}})(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}\Sigma_{zx} = 0.$$

Because $P_{T_{\text{IV}}}$ projects onto $\mathcal{C}\{(\Sigma^{\text{IV}}_{\text{raw}})^{1/2}T_{\text{IV}}^{\top}\}$, this is equivalent to

$$\mathcal{C}\left\{(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}\Sigma_{zx}\right\} \subseteq \mathcal{C}\left\{(\Sigma^{\text{IV}}_{\text{raw}})^{1/2}T_{\text{IV}}^{\top}\right\}.$$

Applying the nonsingular transformation $(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}$ to both spaces gives the equivalent condition

$$\mathcal{C}\left\{(\Sigma^{\text{IV}}_{\text{raw}})^{-1}\Sigma_{zx}\right\} \subseteq \mathcal{C}(T_{\text{IV}}^{\top}).$$

Hence, projection is information-lossless exactly when every efficient raw-IV direction relevant for estimating β lies in the raw-IV subspace retained by T_{IV}. Otherwise, $I_q - P_{T_{\text{IV}}}$ removes a nonzero component of at least one such direction, so the projected system preserves identification but loses efficiency information.

This completes the proof.

□

F. Feasible risk-optimal weighting for the projected GMM system

This appendix provides a feasible implementation of the risk-optimal weighting rule in Section 4. The construction is unified across exact and locally misspecified regimes. We first estimate the

Jacobian and the covariance of the projected first-order scores, and then solve the sample analogue of the K-optimization in (4.3). Exact specification is obtained as the zero-bias special case.

Recall that $J = \mathbb{E}(h_i x_i^\top) = (Q_{\text{IV}}^\top, Q_{\text{SV}}^\top)^\top$, where $h_i = ((\Pi_z x_i)^\top, (\Pi_w^\perp x_i)^\top)^\top$. Its sample counterpart is

$$\widehat{J} := \frac{1}{n} \begin{pmatrix} X^\top P_Z X \\ \\ X^\top P_W^\perp X \end{pmatrix}.$$

The corresponding projected sample moment is

$$\widehat{g}(\theta) := \frac{1}{n} \begin{pmatrix} X^\top P_Z (Y - X\theta) \\ \\ X^\top P_W^\perp (Y - X\theta) \end{pmatrix}.$$

Thus, the feasible procedure uses exactly the same $2p$-dimensional projected moment system as the main-text estimator.

Choose any fixed first-step weight $M_0 \in \mathbb{S}_{++}^{2p}$, for example $M_0 = I_{2p}$, and compute $\widehat{\beta}(M_0) = \operatorname{argmin}_{\theta \in \mathbb{R}^p} \widehat{g}(\theta)^\top M_0 \widehat{g}(\theta)$. Its only role is to provide a consistent preliminary estimator for the plug-in first-order scores.

Let $\widehat{r}_i := y_i - x_i^\top \widehat{\beta}(M_0)$. Define $\widehat{\Sigma}_z := n^{-1} \sum_i z_i z_i^\top$, $\widehat{\Sigma}_{xz} := n^{-1} \sum_i x_i z_i^\top$, and $\widehat{m}_z := n^{-1} \sum_i z_i \widehat{r}_i$. Similarly, define $\widehat{\Sigma}_w := n^{-1} \sum_i w_i w_i^\top$, $\widehat{\Sigma}_{xw} := n^{-1} \sum_i x_i w_i^\top$, and $\widehat{m}_w := n^{-1} \sum_i w_i \widehat{r}_i$.

The sample projected regressors are $\widehat{p}_{z,i} := \widehat{\Sigma}_{xz} \widehat{\Sigma}_z^{-1} z_i$ and $\widehat{v}_i := x_i - \widehat{\Sigma}_{xw} \widehat{\Sigma}_w^{-1} w_i$. To account for estimation of the projection operators, let $\widehat{c}_z := \widehat{\Sigma}_z^{-1} \widehat{m}_z$, $\widehat{c}_w := \widehat{\Sigma}_w^{-1} \widehat{m}_w$, $\widehat{e}_{z,i} := \widehat{r}_i - z_i^\top \widehat{c}_z$, and $\widehat{e}_{w,i} := \widehat{r}_i - w_i^\top \widehat{c}_w$.

The plug-in projected scores are $\widehat{\psi}_{\text{IV},i} := \widehat{p}_{z,i} \widehat{e}_{z,i} + x_i z_i^\top \widehat{c}_z$ and $\widehat{\psi}_{\text{SV},i} := \widehat{v}_i \widehat{e}_{w,i}$. Stacking the

two blocks gives $\widehat{\psi}_i := ((\widehat{\psi}_{\mathrm{IV},i})^\top, (\widehat{\psi}_{\mathrm{SV},i})^\top)^\top$, the feasible analogue of the score in (3.5). The full plug-in scores are required because estimation of the projection operators contributes to the first-order distribution.

Let $\overline{\widehat{\psi}} := n^{-1}\sum_i \widehat{\psi}_i$. We estimate the score covariance by

$$\widehat{\Sigma} := \frac{1}{n}\sum_{i=1}^{n}(\widehat{\psi}_i - \overline{\widehat{\psi}})(\widehat{\psi}_i - \overline{\widehat{\psi}})^\top.$$

Along the local sequence, we write $\mathrm{Var}(\psi_i) \to \Sigma_{\mathrm{miss}}$, where Σ_{miss} denotes the limiting covariance appearing in the local-misspecification analysis. Under exact specification, this reduces to the exact-specification covariance $\Sigma = \mathrm{Var}(\psi_i)$. Thus the same covariance estimator can be used in both regimes.

The feasible risk-optimal sensitivity is obtained by replacing the population quantities in (4.3) with their plug-in estimators:

$$\widehat{K}_{\mathrm{opt}} \in \operatorname*{argmin}_{K:\, K\widehat{J}=I_p} \left\{ \|KD\|_{\mathrm{op}}^2 + \mathrm{tr}(K\widehat{\Sigma}K^\top) \right\}.$$

This directly estimates the first-order sensitivity and avoids a separate optimization over $2p \times 2p$ weighting matrices.

To recover a weighting matrix inducing $\widehat{K}_{\mathrm{opt}}$, let $\widehat{N}_J \in \mathbb{R}^{2p\times p}$ have orthonormal columns spanning $\ker(\widehat{J}^\top)$. The full class of positive-definite weights inducing $\widehat{K}_{\mathrm{opt}}$ is

$$\widehat{\mathcal{M}}_{\mathrm{opt}} = \left\{ \widehat{K}_{\mathrm{opt}}^\top A \widehat{K}_{\mathrm{opt}} + \widehat{N}_J B \widehat{N}_J^\top : A, B \in \mathbb{S}_{++}^p \right\}.$$

Indeed, every $\widehat{M} \in \widehat{\mathcal{M}}_{\mathrm{opt}}$ satisfies $\widehat{J}^\top \widehat{M} \widehat{J} = A$ and $\widehat{J}^\top \widehat{M} = A\widehat{K}_{\mathrm{opt}}$, and hence

$$(\widehat{J}^\top \widehat{M} \widehat{J})^{-1} \widehat{J}^\top \widehat{M} = \widehat{K}_{\mathrm{opt}}.$$

Conversely, the same argument as in the population characterization shows that every positive-definite weight inducing $\widehat{K}_{\mathrm{opt}}$ admits this representation.

For implementation, define $\widehat{P}_{\widehat{J}}^{\perp} := I_{2p} - \widehat{J}(\widehat{J}^{\top}\widehat{J})^{-1}\widehat{J}^{\top}$. For any fixed $\rho > 0$, a convenient representative is

$$\widehat{M}_{\mathrm{opt}}(\rho) := \widehat{K}_{\mathrm{opt}}^{\top}\widehat{K}_{\mathrm{opt}} + \rho\,\widehat{P}_{\widehat{J}}^{\perp}.$$

This matrix belongs to $\mathbb{S}_{++}^{2p}$ and induces exactly $\widehat{K}_{\mathrm{opt}}$. Therefore the corresponding feasible estimator can be written as

$$\widehat{\beta}_{\mathrm{opt}} = \widehat{K}_{\mathrm{opt}}\frac{1}{n}\begin{pmatrix} X^{\top}P_Z Y \\ \\ X^{\top}P_W^{\perp}Y \end{pmatrix} = \operatorname*{argmin}_{\theta\in\mathbb{R}^p} \widehat{g}(\theta)^{\top}\widehat{M}_{\mathrm{opt}}(\rho)\widehat{g}(\theta).$$

The estimator does not depend on the particular $\rho > 0$, or more generally on the choice of representative within $\widehat{\mathcal{M}}_{\mathrm{opt}}$. The role of ρ is therefore distinct from the mixing parameter λ in the restricted WLS path of Section 4.3.

When the local-bias component is absent, $D = 0$, the feasible criterion reduces to $\min_{K:K\widehat{J}=I_p} \mathrm{tr}(K\widehat{\Sigma}K^{\top})$. If $\widehat{\Sigma} \in \mathbb{S}_{++}^{2p}$, its unique solution is

$$\widehat{K}_{\mathrm{eff}} = (\widehat{J}^{\top}\widehat{\Sigma}^{-1}\widehat{J})^{-1}\widehat{J}^{\top}\widehat{\Sigma}^{-1}.$$

The usual two-step efficient-GMM weight $\widehat{\Sigma}^{-1}$ is a canonical positive-definite weight inducing this sensitivity. Thus exact specification is not a separate optimization problem, but the zero-bias special case of the same feasible construction.

Proposition 13 (Consistency of the feasible risk-optimal procedure)**.** Suppose the following conditions hold.

(i) Assumption 4 holds.

(ii) The limiting local score covariance satisfies $\Sigma_{\text{miss}} \in \mathbb{S}^{2p}_{++}$.

(iii) Uniformly along the local sequence, $\sup_n \mathbb{E}\|x_i x_i^\top\|_{\text{op}}^2 < \infty$, $\sup_n \mathbb{E}\|z_i z_i^\top\|_{\text{op}}^2 < \infty$, and $\sup_n \mathbb{E}\|w_i w_i^\top\|_{\text{op}}^2 < \infty$.

Let D be fixed and let K_{opt} denote the unique solution of (4.3), with the limiting local score covariance denoted by Σ_{miss}.

(i) The feasible ingredients are consistent: $\widehat{J} \xrightarrow{\text{p}} J$, $\widehat{\Sigma} \xrightarrow{\text{p}} \Sigma_{\text{miss}}$, and $\widehat{K}_{\text{opt}} \xrightarrow{\text{p}} K_{\text{opt}}$.

(ii) For any fixed $\rho > 0$, define $P_J^\perp := I_{2p} - J(J^\top J)^{-1}J^\top$ and $M_{\text{opt}}(\rho) := K_{\text{opt}}^\top K_{\text{opt}} + \rho P_J^\perp$. Then $\widehat{M}_{\text{opt}}(\rho) \xrightarrow{\text{p}} M_{\text{opt}}(\rho)$, and $\sqrt{n}\{\widehat{\beta}_{\text{opt}} - \beta\} \xrightarrow{\text{d}} \mathcal{N}\big(K_{\text{opt}}\tau,\, K_{\text{opt}}\Sigma_{\text{miss}}K_{\text{opt}}^\top\big)$.

(iii) Under exact specification, $\tau = 0$, $D = 0$, and $\Sigma_{\text{miss}} = \Sigma$. In this case, $K_{\text{opt}} = (J^\top\Sigma^{-1}J)^{-1}J^\top\Sigma^{-1}$, and $\sqrt{n}\{\widehat{\beta}_{\text{opt}} - \beta\} \xrightarrow{\text{d}} \mathcal{N}\big(0,\, (J^\top\Sigma^{-1}J)^{-1}\big)$. Hence the feasible procedure reduces to standard efficient projected GMM when the bias component vanishes.

The feasible construction therefore has two roles. The first step estimates the covariance of the projected first-order scores, whereas the second chooses the sensitivity according to the risk criterion in Section 4. Thus $D \neq 0$ yields a local bias–variance tradeoff, while $D = 0$ reduces the criterion to variance minimization.

F.1 Proof of Proposition 13

Proof. We first establish consistency of the quantities entering the feasible criterion. By Theorem 5, the fixed-weight preliminary estimator satisfies $\widehat{\beta}(M_0) \xrightarrow{\text{p}} \beta$. The law of large numbers

therefore gives $\widehat{\Sigma}_z \xrightarrow{\mathrm{p}} \Sigma_z$, $\widehat{\Sigma}_{xz} \xrightarrow{\mathrm{p}} \Sigma_{xz}$, $\widehat{\Sigma}_w \xrightarrow{\mathrm{p}} \Sigma_w$, and $\widehat{\Sigma}_{xw} \xrightarrow{\mathrm{p}} \Sigma_{xw}$. Because $\Sigma_z \in \mathbb{S}^q_{++}$ and $\Sigma_w \in \mathbb{S}^m_{++}$, the corresponding sample inverses are also consistent. It follows that $\widehat{J} \xrightarrow{\mathrm{p}} J$.

Next consider the plug-in scores. Write $c_z := \Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)$ and $c_w := \Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)$, with $e_{z,i} := \varsigma_i - z_i^\top c_z$ and $e_{w,i} := \varsigma_i - w_i^\top c_w$. Along the local sequence, these population quantities are understood under the corresponding data-generating process. Since $\widehat{\beta}(M_0) \xrightarrow{\mathrm{p}} \beta$, the sample residual moments satisfy $\widehat{m}_z - \mathbb{E}(z_i\varsigma_i) \xrightarrow{\mathrm{p}} 0$ and $\widehat{m}_w - \mathbb{E}(w_i\varsigma_i) \xrightarrow{\mathrm{p}} 0$. Consequently, $\widehat{c}_z - c_z \xrightarrow{\mathrm{p}} 0$ and $\widehat{c}_w - c_w$.

Together with the covariance convergences above and the imposed quadratic-moment conditions, the same Cauchy–Schwarz argument used in the projected first-order expansion yields

$$\frac{1}{n}\sum_{i=1}^{n} \|\widehat{\psi}_i - \psi_i\|_2^2 \xrightarrow{\mathrm{p}} 0.$$

Hence replacing the population scores by their plug-in counterparts is asymptotically negligible for the centered sample covariance.

Under local misspecification, $\mathbb{E}(\psi_i) = \tau/\sqrt{n}$ and $\mathrm{Var}(\psi_i) \to \Sigma_{\mathrm{miss}}$. The triangular-array law of large numbers and the preceding L_2-consistency therefore imply $\widehat{\Sigma} \xrightarrow{\mathrm{p}} \Sigma_{\mathrm{miss}}$. Because $\Sigma_{\mathrm{miss}} \in \mathbb{S}^{2p}_{++}$, continuity of the minimum eigenvalue also gives $\Pr\{\widehat{\Sigma} \in \mathbb{S}^{2p}_{++}\} \to 1$.

It remains to establish consistency of the optimizer. Define $R(K) := \|KD\|_{\mathrm{op}}^2 + \mathrm{tr}(K\Sigma_{\mathrm{miss}}K^\top)$ and $\widehat{R}(K) := \|KD\|_{\mathrm{op}}^2 + \mathrm{tr}(K\widehat{\Sigma}K^\top)$. Since $\Sigma_{\mathrm{miss}} \in \mathbb{S}^{2p}_{++}$, $R(K)$ is strongly convex on $\{K : KJ = I_p\}$, so its minimizer K_{opt} is unique.

Because $K_{\mathrm{opt}}J = I_p$, J has full column rank. Since $\widehat{J} \xrightarrow{\mathrm{p}} J$, $\widehat{J}$ also has full column rank

with probability approaching one. On this event, let

$$\widetilde{K} := K_{\mathrm{opt}} + (I_p - K_{\mathrm{opt}}\widehat{J})(\widehat{J}^\top \widehat{J})^{-1}\widehat{J}^\top.$$

Then $\widetilde{K}\widehat{J} = I_p$, so $\widetilde{K}$ is feasible for the sample problem, and $\widetilde{K} \xrightarrow{\mathrm{p}} K_{\mathrm{opt}}$.

By optimality, $\widehat{R}(\widehat{K}_{\mathrm{opt}}) \le \widehat{R}(\widetilde{K})$. Moreover, with probability approaching one, $\lambda_{\min}(\widehat{\Sigma})$ is bounded away from zero, and hence $\widehat{R}(K) \ge \lambda_{\min}(\widehat{\Sigma})\|K\|_F^2$. Since $\widehat{R}(\widetilde{K}) = O_p(1)$, we obtain $\|\widehat{K}_{\mathrm{opt}}\|_F = O_p(1)$.

Consider any convergent subsequence of $\widehat{K}_{\mathrm{opt}}$, with limit $\overline{K}$. The constraint $\widehat{K}_{\mathrm{opt}}\widehat{J} = I_p$ and $\widehat{J} \xrightarrow{\mathrm{p}} J$ imply $\overline{K}J = I_p$. Together with $\widehat{\Sigma} \xrightarrow{\mathrm{p}} \Sigma_{\mathrm{miss}}$ and the preceding boundedness, optimality implies $R(\overline{K}) \le R(K_{\mathrm{opt}})$. Uniqueness of the population minimizer therefore gives $\overline{K} = K_{\mathrm{opt}}$. Hence $\widehat{K}_{\mathrm{opt}} \xrightarrow{\mathrm{p}} K_{\mathrm{opt}}$.

Since $\widehat{J} \xrightarrow{\mathrm{p}} J$ and both matrices have full column rank with probability approaching one,

$$\widehat{P}_{\widehat{J}}^{\perp} = I_{2p} - \widehat{J}(\widehat{J}^\top \widehat{J})^{-1}\widehat{J}^\top \xrightarrow{\mathrm{p}} P_J^{\perp}.$$

Together with $\widehat{K}_{\mathrm{opt}} \xrightarrow{\mathrm{p}} K_{\mathrm{opt}}$, the continuous mapping theorem gives

$$\widehat{M}_{\mathrm{opt}}(\rho) = \widehat{K}_{\mathrm{opt}}^\top \widehat{K}_{\mathrm{opt}} + \rho\widehat{P}_{\widehat{J}}^{\perp} \xrightarrow{\mathrm{p}} K_{\mathrm{opt}}^\top K_{\mathrm{opt}} + \rho P_J^{\perp} = M_{\mathrm{opt}}(\rho).$$

This proves the claimed convergence of the feasible weighting-matrix representative, without requiring a separate equation label.

Finally, since $\widehat{K}_{\mathrm{opt}}\widehat{J} = I_p$, the definition of $\widehat{\beta}_{\mathrm{opt}}$ gives

$$\sqrt{n}\{\widehat{\beta}_{\mathrm{opt}} - \beta\} = \widehat{K}_{\mathrm{opt}}\sqrt{n}\,\widehat{g}(\beta).$$

The projected-moment expansion underlying Theorem 5 gives

$$\sqrt{n}\,\widehat{g}(\beta) \xrightarrow{\mathrm{d}} \mathcal{N}(\tau, \Sigma_{\mathrm{miss}}).$$

Combining this result with $\widehat{K}_{\mathrm{opt}} \xrightarrow{\mathrm{p}} K_{\mathrm{opt}}$ and Slutsky's theorem yields the stated local asymptotic distribution.

When $D = 0$, the population criterion reduces to $\mathrm{tr}(K\Sigma K^\top)$ under exact specification. A standard Lagrange-multiplier calculation gives $K_{\mathrm{opt}} = (J^\top \Sigma^{-1} J)^{-1} J^\top \Sigma^{-1}$, and substitution gives the efficient asymptotic variance $(J^\top \Sigma^{-1} J)^{-1}$. The usual efficient-GMM weight Σ^{-1} is a canonical positive-definite weight inducing this sensitivity. This is exactly the efficient-GMM special case described in the main text.

□